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Second Cycle, 30 credits

Concept Development of a New Battery Suspension for BEV Trucks

For Scania’s Special Order Department

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Concept Development of a New Battery Suspension for BEV Trucks

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Abstract

This thesis focuses on developing a new battery suspension concept for Scania's electric Basic Trucks. The need for this development arises from the limitations of the current battery suspension design in Scania's existing generation of battery electric vehicles (BEVs), which restricts the installation of various required bodywork. These bodyworks often interfere with the current suspension system, making it incompatible with multiple customer applications.

To address this issue, a pre-study was conducted to understand the rationale behind the existing suspension design. This included a market analysis and the identification of customer needs specific to Basic Trucks. Based on the findings, the Quality Function Deployment (QFD) tool was used to derive the key requirements for the concept development.

Several methods, including Function Means Tree (FMT) and the Pugh Matrix, were applied to generate and evaluate concept alternatives. One concept was selected and further developed. Its feasibility was validated through Generative Assembly Structural Analyses using Catia V5.

The project concluded that a redesigned battery suspension system is essential if Scania intends to meet more special customer demands for electric Basic Trucks. The analysis demonstrated that a one-sided mounted battery suspension using the MP10 propulsion battery is feasible for installation. Further development and refinement of this concept are necessary to provide a viable alternative battery suspension solution in the future.

Keywords:

Battery suspension, BEV, Basic-Trucks, concept development, Catia V5

Sammanfattning

Detta examensarbete fokuserar på att utveckla ett nytt koncept för batteriupphängning till Scanias elektriska Basic Trucks. Behovet av denna utveckling uppstår på grund av begränsningarna i den nuvarande batteriupphängningsdesignen i Scanias befintliga generation av batterielektriska fordon (BEV), vilken försvårar installationen av olika kundanpassade påbyggnationer. Dessa påbyggnationer kolliderar ofta med den nuvarande upphängningen, vilket gör den inkompatibel med flera applikationer.

För att hantera problemet genomfördes en förstudie med syfte att förstå bakgrunden till den existerande designen. Studien inkluderade en marknadsanalys samt en kartläggning av kundbehov för Basic Trucks. Utifrån dessa insikter användes verktyget Quality Function Deployment (QFD) för att ta fram centrala krav för konceptutvecklingen.

Flera metoder, såsom Function Means Tree (FMT) och Pugh-matrisen, användes för att generera och utvärdera konceptalternativ. Ett lovande koncept valdes ut och vidareutvecklades. Dess genomförbarhet verifierades genom generativa strukturanalyser i Catia V5.

Projektet drog slutsatsen att ett omdesignat batterifästsystem är avgörande om Scania ska kunna möta fler särskilda kundkrav för elektriska Basic Trucks. Analysen visade att ett ensidigt monterat batterifäste med framdrivningsbatteriet MP10 är genomförbart att installera. Vidare utveckling och förfining av detta koncept är nödvändigt för att i framtiden kunna erbjuda en fungerande alternativ lösning för batterifäste.

Nyckelord:

Batteriupphängning, BEV, Basic-lastbilar, konceptutveckling, Catia V5

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Furthermore, we would like to thank everyone at KTH who contributed to our thesis work, with special appreciation to our supervisor Conrad Luttropp. His guidance helped us focus on the key aspects necessary to achieve meaningful results. We also extend our thanks to all the professors at KTH, whose teaching and preparation over the past two years laid the foundation for this achievement.

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List of Acronyms and Abbreviations

A-order	Order of Standard Trucks
A-Truck(s)	Articulated Truck(s)
BEV	Battery Electric Vehicle
B-Truck(s)	Basic Truck(s)
CEU	Central Electric Unit
ICE	Internal Combustion Engine
LFP	Lithium-Ferro-Phosphate
kW	Kilowatt
kWh	Kilowatt hours
LHS	Left-Hand-Side
MP	Mega Pack
MPa	Megapascal
MS	Mega Structure
NCA	Nickel Cobalt Aluminium
NMC	Nickel Manganese Cobalt
RHS	Right-Hand-Side
S-order	Special order department
VCB	Voltage Class B

1. Introduction

This Section will give an overview of the project which will be conducted in this work. Therefore, the background of the topic will be explained, and the purpose and definition will be clarified. The Section will finish with the explanation of which research methodologies were applied and for which purpose.

1.1 Background

Scania AB is a worldwide provider of transport solutions, based in Södertälje, Sweden. Over the last few decades, the industry was dominated by transportation solutions, which were powered by combustion engines, so-called ICE trucks. Due to the fact that the transport industry is a major contributor to CO_2 emissions, the industry aims to become more sustainable. Thereby, one approach is to lower the produced emissions during operation, which is why Battery Electric Vehicles (BEV) were introduced (Scania AB, 2025b).

Scania introduced its first BEV truck back in 2020, where a BEV truck was designed based on the chassis of an internal combustion engine (ICE) truck. Up to six battery packs with a total installed capacity of 300 kWh were attached on the LHS and RHS of the chassis, between the front and rear axles. Each battery pack was attached with brackets to the chassis, which was similar to how fuel tanks on an ICE truck are attached (Scania AB, 2020).

With the product launch of the 3rd generation of BEV trucks in 2024, a new battery suspension was introduced. Thereby, the need arose to fit batteries with higher capacities to keep up with the customer demands. Instead of 300 kWh, the new BEV truck can be equipped with an installed capacity of up to 624 kWh, which results in bigger and heavier battery packs. The Mega Structure (MS) is mounted at a similar position to the chassis as in the past, on the LHS and RHS of the frame between the front and rear axle. This position of the MS is defined and cannot be changed currently, which results in issues for Scania to fulfil different special customer applications. The department within Scania, which is responsible for solving special customer requests (so-called S-order), received several requests to move the MS in its position or to remove one battery pack to gain free space. This need results because in the past, components, such as fuel tanks, were available in different sizes and flexible in their position on the chassis, which allowed a certain flexibility. With the current design of the MS, this flexibility is not ensured.

To enable the different special customer requests, it needs to be able to move the position of the MS, which would result in more flexibility for Scania to fulfil the special customer needs.

1.2 Purpose and Definition

The purpose of this study was to increase the number of special customer applications, Scania can offer their BEV trucks to. This project investigated the current design of the MS and developed ideas for a new conceptual design of a battery suspension, which addresses the need for more flexibility in the position of the batteries. As a result, a concept of a new battery suspension was developed. The work was conducted within the S-Order department within Scania, to find a solution which allows them to solve different special requests regarding the position of the battery suspension.

1.2.1 Project Missions

The project aimed to address the special customer requests for more flexibility in the position of the batteries, to be able to mount the required bodywork to B-Trucks

1.2.2 Project Objectives

The study objective was to analyse the current design of the battery suspension and fully understand the strengths and weaknesses of the design. Furthermore, a competitor analysis was conducted to identify available solutions on the market and to compare them to Scania, to identify similarities and differences. Moreover, the special customer request regarding bodywork was analysed to identify where the free space on the truck is required. Lastly, before starting with the concept generation, an analysis was conducted to distinguish areas on the truck, where propulsion batteries can be placed.

Based on the study's requirements and the insights gathered, several concepts were developed to meet the customer's needs. From these concepts, one was selected for further development, evaluation, and optimisation. This led to the creation of a final design. The questions that were answered during the study were:

- Which customer applications and project targets were in focus during the development of the MS?
- What are the strengths and weaknesses of the MS?
- Which customer applications are not applicable with the position of the MS?
- Why is the battery suspension not movable forward or backwards today?
- What does the competition offer, and how flexible are their battery suspensions?
- Which special customer applications are currently not applicable with the MS?
- Where is space available on the truck to install propulsion batteries?

1.2.3 Project Deliverables

The project deliverables were:

- Analysing the current design of the MS and understanding the reasons behind the design.
- A market analysis of how competitors design their battery suspensions and how they differ towards Scania's.
- Identifying space for the installation of propulsion batteries.
- Concept generation for a new battery suspension.
- Concept evaluation and further developing the chosen concept.
- Analyses which consider the stress in the frame of the truck.
- Presenting a final concept which includes a new battery suspension, which enables more special customer applications.
- The final concept is developed for the special order department within Scania, which seeks a concept which can be implemented quickly and cost-effectively.

1.2.4 Project Delimitations

Besides the described objectives and deliverables of the project, the following aspects are the delimitations of the project, which were not part of the project:

- Developing new propulsion batteries
    - To enable the development of a concept which is feasible to implement, the developed concept will use the propulsion batteries of Scania, which are currently on the market.
- No prototype testing
    - Due to the complexity of a battery suspension and the involved costs when producing prototypes, no prototypes will be produced during the project.
- No physical endurance or fatigue testing
    - The conducted simulations will be limited to analyses within Catia V5, which is why no physical endurance or fatigue tests will be conducted.
- Cost analysis
    - No cost analysis will be conducted for the developed concept of a battery suspension. The focus was on finding a solution for the identified problem.
- Implementation into production
    - The implementation of how and where the developed concept can be produced will not be considered in the project.

1.2.5 Project Stakeholders

As a start of the project, a stakeholder analysis was conducted to receive an overview of the involved stakeholders in this project, which can be seen in Figure 1.

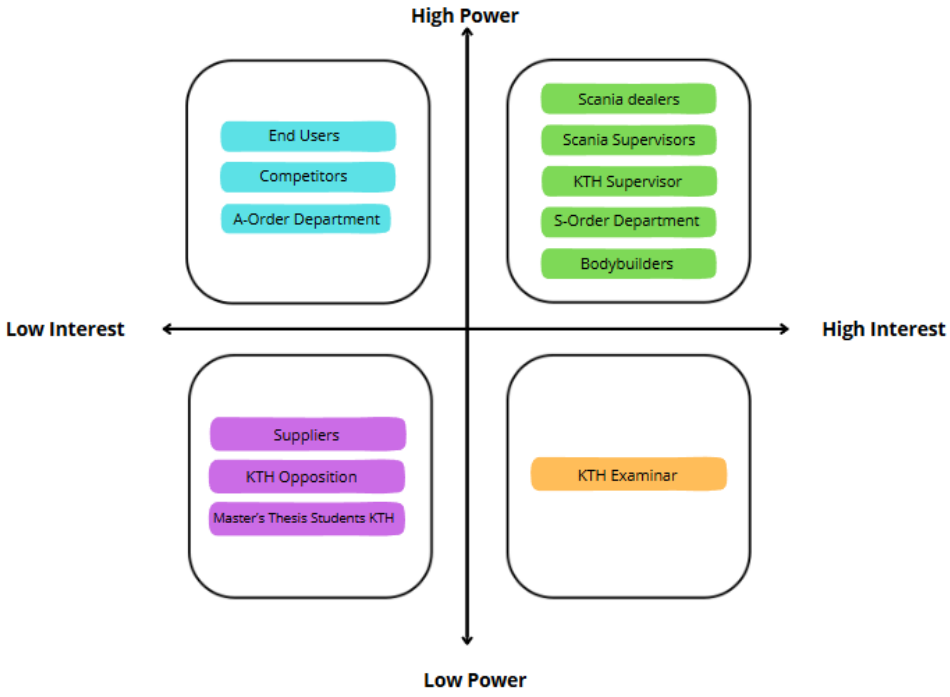


Figure 1: Stakeholder Analysis

Low interest - high power

The stakeholders with high power but low interest in this project include end users, competitors of Scania and the A-order department.

End users have high power because they are the ones deciding to buy the product. Their needs and applications define the requirements and specifications for the design process. They have an interest in the solution, but if Scania can not deliver a solution for their needs, they have the alternative to buy from the competition. This results in a lower interest, compared to other stakeholders.

The competitors of Scania have a high power, because if they can offer products for the customer applications, Scania cannot currently address, they have a competitive advantage. Moreover, the competition has no interest in a successful implementation of this project because it threatens their competitive advantage.

Furthermore, the A-order department has significant power in the project because they are the ones who developed the MS and have in-depth knowledge of its limitations, weaknesses, and strengths, which will have a major impact on the project. However, their interest will be low since the project is intended for the S-order department and not the A-order department.

High power - high interest

Key stakeholders who have high power and interest in the project include Scania dealers, Scania supervisors, KTH supervisors, the S-order department, and bodybuilders. Scania dealers will have a central role, as they are the ones selling the products and having direct contact with the end users. They can provide valuable feedback on market trends, competitors, and the end users' needs, which will help the company adapt its products or strategies. Furthermore, their interest is high because they lose customers if they cannot meet customer requests, but the competition can.

The Scania and KTH supervisors will also have a high power during the project, because they are in close contact with the project team during the project. With their experience and decision-making skills, they navigate and support the project team to move forward. Moreover, the KTH supervisor is highly interested in accomplishing a good academic research paper, which fulfils the requirements. Compared to that, the Scania supervisors are interested in the result of the project, which aims to benefit their offering in the future.

The S-order department is one of the key stakeholders since they aim to investigate and understand the current design of the MS. Since this department handles special orders for their customers, the project will be beneficial to them. They will be the ones most affected by the results, which results in a high interest.

Additionally, the bodybuilders have a big impact on the project since they are responsible for the bodywork on a B-Truck. As a result, they have a high power in the project, because they know the customer applications and need to be able to assemble them on the Scania trucks. Moreover, their interest is high as well, because they earn money in the moment, Scania receives an order for a B-Truck.

High interest - low power

The one who has high interest but low power in this project is the KTH examiner, as they are responsible for reviewing and grading the project. Their interest mainly lies in understanding the objectives, methodology, and results of the project. At the same time, they are also invested in the project's success and its learning outcomes. They do not participate in decision-making and cannot change the direction of the project, which results in low power.

Low power - low interest

The stakeholders with limited power or interest in the project include suppliers, KTH opposition and other master thesis students at KTH. The supplier's responsibility is to provide the necessary materials and components, but they are not involved in making decisions in the process or providing directions in the project. They are not invested in the project´s success or failure because of the fact that suppliers usually operate at the transactional level.

The KTH opposition oversees the project's review and critique from an academic perspective. Their role is limited to evaluating and providing advice, not making decisions or guiding the direction of the project. The same applies to other master's thesis students at KTH, as they are focused on their projects and are not involved in the execution, strategy, or decisions made within the project. They may have some interest in the project, mainly to see how other students are handling similar problems. However, their interest is usually academic and does not have any direct effect on the project's progress.

1.2.6 Project Planning

A work breakdown structure (WBS) was used to create a clear overview and break down the project into smaller, manageable tasks see Appendix A.

1.3 Research Methodology

In this chapter, an overview of the applied methods in the master's thesis will be given to describe how and why the methods were applied.

Stakeholder Analysis

As an initial step in the project, a stakeholder diagram was created to identify and clarify the actors who are affected by or have an interest in the project. The main reason for creating the stakeholder diagram was to get a clear picture of the project’s context and to ensure that relevant perspectives and needs were considered throughout the process. By visualising the relationships between stakeholders, it was possible to identify requirements, interests, and conflicts early on. This allowed for better planning and gave a clearer understanding of stakeholder priorities.

Work Breakdown Structure

After implementing the stakeholder diagram, a work breakdown structure was created. The purpose of this was to break down the overall project into smaller, more manageable tasks or components. This makes it easier to understand the full scope of the work, assign responsibilities, and estimate the time and resources needed for each part. In addition, it supports better planning, progress tracking, and risk identification early in the process, which increases the chances of completing the project on time and within scope. The breakdown can be found in Appendix A.

Discussions & Interviews

To further explore the project topic, discussions and interviews were held with key individuals who had been part of creating the MS. These appointments helped clarify the intentions behind the battery suspension design and gave insight into whether other design ideas had been considered. The input also helped in identifying potential limitations and advantages of the existing solution, which served as helpful guidance in the continued development. In addition, the participants pointed to other people with relevant experience who could provide further input. These conversations were highly beneficial, as they highlighted challenges that had already been encountered and addressed in previous development. This made it possible to avoid repeating earlier mistakes and to build upon existing knowledge.

Market Analysis

A market analysis was conducted as an important part of the project methodology. It provided a clear overview of the current market conditions, competitors, and customer needs. This analysis helped identify potential opportunities and challenges related to the concept development. Understanding the market landscape made it possible to align the project goals with real demand and trends, reducing the risk of developing a product that does not meet user expectations.

Quality Function Deployment

The Quality Function Deployment (QFD) was used to deepen the understanding of customer needs and how to meet them effectively. This method served as a foundation for the design decisions, helping to focus on the most important aspects to consider. QFD is crucial because it translates customer requirements into clear design priorities. Overall, QFD guided the development process by ensuring that the project focused on delivering features that truly matter to customers while maintaining a competitive edge in the market. The QFD can be found in Appendix B.

Function Means Tree

The Function Means Tree (FMT) method helped break down the overall function of the product and identify possible means to achieve each one. By using FMT, it became easier to compare different ideas systematically and understand how well each one addressed the functional needs of the design. It also provided a clear structure for exploring alternative solutions and analysing their strengths and weaknesses. Based on the identified functions and means, various combinations were generated to explore how different ideas could be merged. In total, eight concept proposals were formulated. Each of these was evaluated to determine its potential for further development. The final FMT can be seen in Appendix C.

Pugh Matrix

To support the concept selection process further, a Pugh Matrix was used. This method made it easier to compare the different concepts against each other using a structured and consistent approach. By selecting a reference concept and scoring the others relative to the customer requirements of the QFD tool, the team was able to make more objective and balanced decisions. The use of the Pugh Matrix also helped highlight the strengths and weaknesses of each concept, making it clearer which ideas had the most potential and which ones needed further improvement or could be ruled out.

Catia V5

After selecting the final concept using the Pugh Matrix, the new battery suspension design was created in the Catia V5 program and integrated into the truck frame. This step was crucial for visualising how the design would function in a realistic environment and for assessing its spatial fit with existing components. Creating a detailed 3D model allowed the examination of dimensions, making it possible to detect and prevent any potential interference or collisions with other parts early in the process. This approach helped to confirm that the new suspension system would be compatible with the overall truck structure and ensured that the design met space constraints and functional requirements.

The last step was to verify that the new battery suspension design can reliably support the battery without failure or damage. Stress analyses were conducted using Catia Generative Assembly Structural Analysis. These analyses assessed whether the frame can withstand the expected loads and stresses during operation, ensuring that no parts would deform excessively or break. By simulating different stress scenarios, potential weak points could be identified and addressed early, which helps improve the overall safety and durability of the suspension system.

The chapter will provide a detailed explanation of the MS components and their functionalities. It will also cover the key components and electronics that are connected to the MS. Furthermore, the chapter will include a market analysis to gain insight into how competitors designed their battery suspension. The aim thereby is to distinguish the similarities and differences compared to Scania. Moreover, important bodyworks, which are mounted on B-Trucks, will be analysed to identify their position and conclude requirements for the design of the battery suspension concept. Lastly, the identification of free space on the truck for the installation of propulsion batteries will be conducted. With the mentioned analyses, the target is to identify valuable insights to define the requirements for the concept development.

2.1 Scania Electric Trucks

Scania's newest generation of BEV trucks are available as B-Trucks & A-Trucks. B-Trucks carry their cargo, such as fire engines, bin lorries or tipper trucks. B-Trucks are getting equipped with bodywork from bodybuilders to meet different customer needs. These trucks are commonly used in urban applications, where shorter driving distances are required (Scania BIBAT, 2016b). A-Trucks, in comparison, are always equipped with a fifth wheel, where a trailer gets connected for transporting cargo. These trucks are used for long-distance transportation of heavy-haulage or normal (Scania BIBAT, 2016c).

The trucks are available in different wheel configurations, which vary in the number of axles and driven axles. In combination with the different wheel configurations, axle distances of 3250 mm – 6350 mm result (Scania BIBAT, 2024). The axle distance describes the distance between the leading front axle and the leading rear axle (Scania BIBAT, 2016a).

Table 1: Wheel Configurations BEV Trucks

B-Trucks (Tyres x driven tyres)	**A-Trucks** (Tyres x driven tyres)
B 4x2	A 4x2
B 6x2	A 6x2
B 6x4	A 6x4
B 6x2*4	

The chassis of a truck is the frame, which runs from the front to the rear of the truck. The frame consists of two frame members, which are available in different thicknesses, to handle different loads. In total, Scania offers five different types of frames. Three of them have single frame members per side, and the dimensions differ, which results in different strengths. Furthermore, two frames are available, which have double frame members per side. The so-called double frames differ in the thickness of the second frame member, why two different strengths result. The double frame offers the highest frame strength and is used for heavy applications, which are defined by heavy loads and uneven roads (Scania BIBAT, 2016e).

The frame members of the frame are connected by crossmembers, which are located at multiple positions between the frame to give the frame strength and stiffness. The frame has a hole pattern, based on the truck configuration, which allows for mounting components to the frame. At the front, the distance between the frame members is bigger, to have more space to install components, which results in a V-shape (see Figure 2). Above the V-shape, the driver cab is installed, which contains underneath several components such as electrics or cooling systems and the front axle (Scania BIBAT, 2016e). Due to the V-shape, the frame of a BEV truck has a double bend, achieving an earlier parallel straight running frame behind the driver's cab. This design was chosen to have more space to install battery packs on the frame, which requires a straight running frame without an angle. The battery suspension (MS) of the BEV trucks, starts after the second bend, where the frame runs parallel. At the same position between the frame, the powertrain is installed, which includes the electric motor and gearbox. Depending on the axle distance, the distance between the end of the MS and the rear axle can vary (Järlstig, 2024).

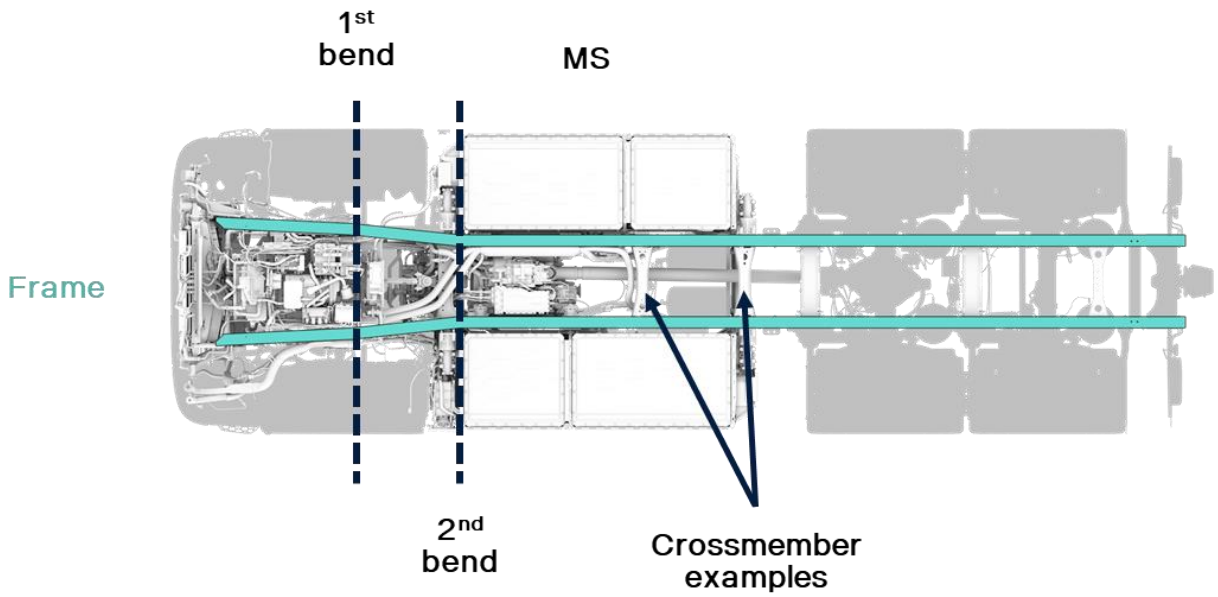


Figure 2: BEV Truck Layout

2.1.1 Mega Structure

While designing a new battery suspension for the next generation of BEV trucks, Scania was focusing on an A 4x2, which is used for long-haulage applications. This application and truck configuration was chosen because it accounted for the highest share of sold trucks when the MS was developed (Forsell, 2025; Kvaldén, 2025). When developing an articulated BEV truck, the range per full charge of the truck is of high importance. Truck drivers in Europe are allowed to drive 4,5 hours without a break (European Union, 2024). As a result, the design of the new battery suspension focused on enabling the storage of the highest possible battery capacity in the most compact design, to drive 4,5 hours without a needed break. A compact design is required to prevent the vehicle from becoming too long, as that would make it impossible to meet legal requirements for the turning radius (Kvaldén, 2025). In order to achieve this, Scania has introduced the MS, which serves as a housing unit and suspension for the propulsion batteries in Scania's BEV trucks, providing protection and structural support, as shown in Figure 3. At the same time, the MS supports itself without any additional frame reinforcements, because of its stiffness and rigidity (Borreani, 2025a).

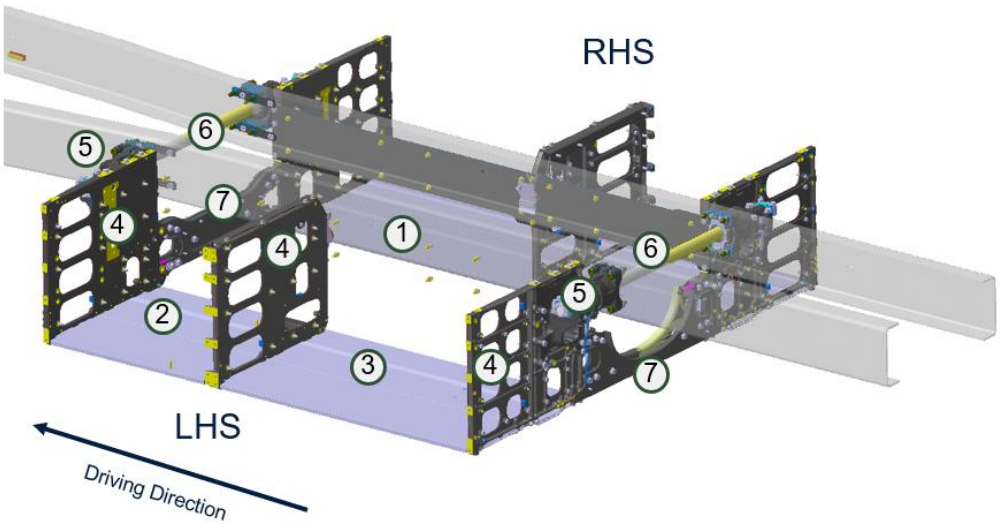


Figure 3: Mega Structure

The MS is designed with separate battery Sections on the LHS and RHS of the truck, with the frame (1) running between them, as it is shown in Figure 3. The two Sections are symmetrical and contain the same components. The only difference is that one of them has been rotated 180 degrees (Kvaldén, 2025).

When observing the LHS, it consists of two different container sizes, where the propulsion batteries MP10 (2) and MP20 (3) are positioned, which stands for Mega Pack. At the side of the batteries, bulkheads (4) are attached. The batteries, in combination with the bulkheads, create a stable connection, which is stiff and hard to twist (Borreani, 2025a). Connected to the bulkheads at the front and end of the MS, a clamping bracket is mounted. The clamping bracket (5) is connected to the bulkhead and frame and clamps a beam. It contains several layers, such as bearings or bushings, which are responsible for reducing the vibrations within the suspension. Two beams (6) run at the front and back of the MS from the LHS to the RHS through the frame and are connected with another clamping bracket on the RHS to the frame. This design was chosen because the beam transfers the stress between the LHS and RHS of the frame (Borreani, 2025a). As a result, the centre of gravity of the MS sits between the two frame members, which is why the frame is balanced and is not twisted to one side. This is of high relevance, because the described installation, combined with the propulsion batteries, has a total weight of above 3,5 tonnes. Furthermore, at the front and bottom of the MS, below the beam, is another connection between the LHS and RHS with a beam (7). The two beams prevent a twisting of the MS below the frame. The beams are connected at the bottom of the bulkheads, where an interface for the installation is present (Borreani, 2025a).

The presented MS with two propulsion batteries per side, described in the following, has a total length of approximately 2600 mm in the driving direction, a width of 2400 mm, and a height of 640 mm. With these dimensions, the MS achieves a total battery capacity of 624 kWh.

2.1.2 Propulsion Batteries

The BEV trucks of Scania are equipped with their own developed and produced battery packs. In cooperation with the battery cell manufacturer Northvolt, Scania developed its own NMC propulsion batteries. Scania receives the battery cells from Northvolt and assembles them into modules in the production in Södertälje. Out of these modules, battery packs are assembled. The software and hardware of these battery packs are developed by Scania itself (Scania AB, 2025a).

Scania currently produces two different Battery packs, MP10 and MP20. The battery packs differ in capacity, weight and dimension. The MP10 and MP20 are mounted in the MS on the LHS and RHS of the chassis, behind the driver's cab (Scania BIBAT, 2023). Furthermore, Scania communicated the release of the MP12 battery for 2026, which will be placed below the driver’s cab. With this implementation, the capacity of the truck can be further increased in the future. Due to the chosen name, a capacity of up to 120 kWh can be expected (Scania AB, 2025c).

Table 2: Propulsion Battery Attributes Scania

	MP10	MP20
Capacity (installed/usable)	104 kWh/100 kWh	208 kWh / 200 kWh
Modules	10x18	20x18
Dimension	890x880x640 mm	1550x880x640 mm
Weight	600 kg	1200 kg
Position	LHS + RHS on the chassis	LHS + RHS on the chassis

2.1.3 Connected components to the Mega Structure

Besides the MS and the propulsion batteries, several other components are connected to the MS. The following components are needed for the functionality of the MS and to fulfil the system.

Central electric unit (CEU)

The central electric unit is one of the main components of a BEV truck because all VCB cables are connected to it. Due to this, the CEU is connected with the propulsion batteries, the PTO unit, inverters, the charging port and the powertrain. The CEU is mounted at the front bottom of the MS and is therefore centrally positioned on the truck (Järlstig, 2024).

Voltage Class B (VCB)

The Voltage Class B (VCB) describes the electrical system with a voltage range of 30 V to 1000 V AC and 60 V to 1500 V DC. The BEV trucks of Scania use a 700 V system, which is why VCB cables are used. All the cables are connected to the CEU, from where the VCB cables continue to the components which are part of the high voltage system. The VCB cables run from the CEU to the MS, where they end in the charging port and junction boxes, which will be described in the following (Järlstig, 2024).

Junction Box and Primary Charging Box

The battery junction box is connected to the propulsion batteries and contains the plus and minus poles of the batteries. The VCB cables are connected to the junction box and go from there to the CEU. The position of the junction box depends on the propulsion battery. The MP10 has four layers of battery modules, the top two layers contain three battery modules, and the bottom two layers contain two battery modules. Due to this, the battery junction box is attached at the side of the bottom, below the additional battery modules of the upper battery layers. Due to this, the dimensions of the MP10 are not increasing, due to the junction box. The MP20 has four layers, and each contains five battery modules. As a result, the battery junction box is attached to the right side of the battery, increasing the total length of the battery (Järlstig, 2024).

To charge the propulsion batteries, the primary charging box is positioned on the left side of the truck behind the front wheel, and the charging power for the batteries can reach up to 375 kW. The charging box is connected with VCB cables to the CEU.

Inverters

The BEV trucks are equipped with inverters, which invert the DC power from the propulsion batteries into AC. The inverters are required so that the AC electric motors, such as the powertrain and the PTO, can be operated. Depending on the selected powertrain and if a PTO is equipped, the position and number of the inverters differ. The inverters are attached below the driver's cab or on top of the powertrain (Järlstig, 2024).

Powertrain and Power Take-Off (PTO)

The powertrain is located between the MS, as it is attached between the frame. The powertrain includes the electric motor and gearbox of the truck. The BEV trucks of Scania can be equipped with three different powertrains. The powertrains differ in power, the number of electric machines, the number of gears and weight. As a result, the dimensions of the powertrains increase the higher the power of the truck is. The powertrains are connected with a propeller shaft to the rear axle. The maximum performance of the powertrain is 450 kW. Furthermore, the powertrain is connected with VCB cables to inverters, from where VCB cables are attached to the CEU (Järlstig, 2024).

Moreover, for applications where additional equipment on the truck needs to be powered electrically or hydraulically, the truck will be equipped with Power Take-Offs (PTO’s). This need occurs if temperature, rotational movement or longitudinal movements are required. Three different PTO’s are available, with two attached to the powertrain and one placed below the driver's cab (Järlstig, 2024).

Cooling system

The cooling system is one of the most important components for the propulsion batteries to operate perfectly. The batteries are designed to operate within specific temperature limits, and the cooling system helps maintain a stable temperature within the battery pack. The cooling system unit is placed below the driver's cab, and from there it is connected to the propulsion batteries and the charging box (Järlstig, 2024).

2.2 Market Analysis

This Section aims to gain information about the main competitors of Scania when it comes to BEV trucks. Therefore, the offering of the competition will be analysed to gain information about their battery suspensions and comparing them towards the MS of Scania. During the analysis, a focus will be placed on the battery capacities, battery types, drive units, the mounting of the batteries to the truck and the different positions of the batteries. To gain the information, an internet investigation was conducted.

2.2.1 Mercedes-Benz

The eActros 600 is the newest BEV truck of Mercedes-Benz and went into series production in November 2024 (Daimler Truck, 2024). The truck is configurable as a B-Truck, with two different axle distances and as an A-Truck, with one axle distance.

The truck is equipped with LFP batteries with a nominal capacity of 621 kWh. The usable battery capacity is 600 kWh and is generated by three battery packs. The propulsion batteries are installed from the LHS to the RHS underneath the frame, see Figure 4. Due to the positioning below the frame, the height of the battery packs is lower compared to Scania. Therefore, the length of the battery is longer. The battery packs are attached with brackets to the side of the chassis (Mercedes-Benz Trucks, 2025). The placement below the frame is possible for Mercedes-Benz to achieve, because they have no powertrain and propeller shaft between the frame at the front, due to the use of an electric axle. As a result, at the position of the battery packs, they have no components which collide with them and can use the free space for the propulsion batteries.

The eActros 600 is equipped with an 800-volt electric axle with two electric motors and a four-speed transmission. The electric axle sits between the driven rear axle. The e-axle produces 400 kW continuously and 600 kW as peak performance (Daimler Truck, 2023).

Figure 4 gives an overview of how Mercedes-Benz solved the packaging of the required components for BEV trucks. The cooling unit of the truck sits underneath the driver's cab, in the position where the engine of an ICE truck is originally positioned. The CEU is located under the driver's cab as well. The charging port sits on the LHS above the batteries. Furthermore, converters/inverters are located above the second battery pack, connected with the CEU at the front and the electric rear axle at the back.

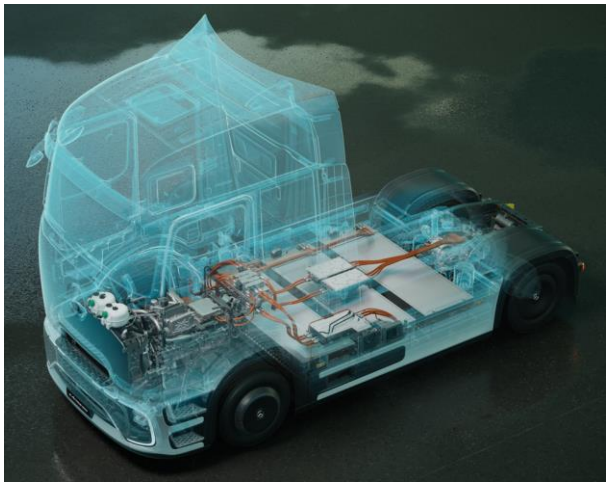


Figure 4: Mercedes-Benz eActros 600 (Mercedes-Benz Trucks, 2025)

2.2.2 Volvo

Volvo's latest electric trucks series are the FH Electric and FM Electric, which are available as B- and A-Trucks. Figure 5 shows an A-Truck with 6 propulsion batteries, which is valid for an FH and FM Electric.

The FH Electric can be equipped with 4-6 battery packs of NCA propulsion batteries, which provide a nominal capacity of 360-540 kWh. With the amount of installed batteries increasing, the usable battery capacity is 65-70%. Each battery pack has a nominal capacity of 90 kWh and is shaped like a cube. The battery packs are attached on the LHS and RHS of the chassis between the front and rear axles, meaning there are six positions to mount the battery packs to the frame. This results in flexibility to fulfil the different customer applications. The LHS and RHS can be equipped with two or three battery packs, in positions 1 + 2 and 1 + 2 + 3, see Figure 5.

Compared to the FH Electric, the FM Electric can be quipped with 2-6 battery packs, which result in a nominal capacity of 180-540 kWh. The mentioned battery configurations above are applicable, and the following two are added. A truck with two batteries has two battery packs at positions 1 + 2 on the LHS and no batteries on the RHS. When 3 battery packs are selected, one battery pack on the RHS in position 2 is added. With 4, 5 and 6 battery packs, the possible configurations are equal to the FH Electric (Volvo Trucks, 2024b).

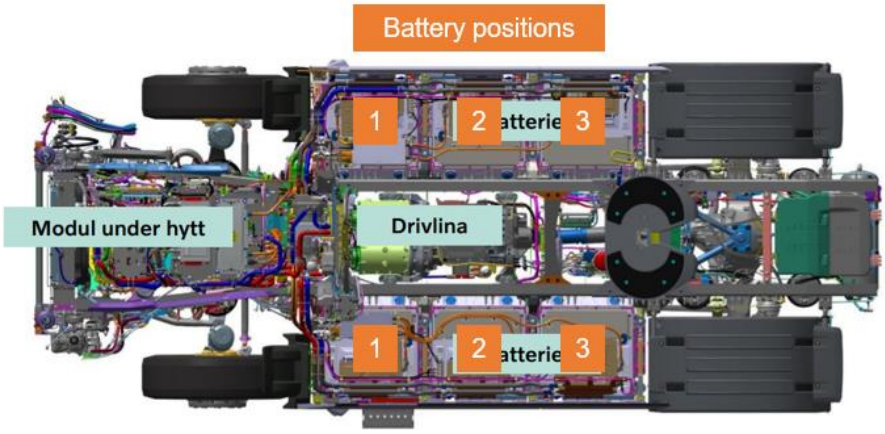


Figure 5: Volvo FH Electric Articulate 6 Battery Packs (Volvo Trucks, 2023)

The electric motor and gearbox are at both models in the same position as with the Scania. It is attached between the frame between the battery packs. With a propeller shaft, the rear axle is driven. The truck can be configured with two or three electric motors, which provide a continuous power of 330-490 kW.

Based on Figure 5 the additional required components for a BEV truck, such as the cooling unit or CEU, are positioned at the front of the truck, below the driver's cab.

Volvo communicated in September 2024 that in 2025, a new FH Electric will be launched, which will provide higher battery capacities. This will be accomplished due to the newly developed electric axle, which fits the electric motor and rear axle into the rear axle. As a result, more space for battery packs is available, because the space between the frame, where the electric motor and transmission are currently positioned, becomes available for battery packs (Volvo Trucks, 2024a).

2.2.3 DAF

The latest BEV trucks of DAF are the DAF XB, XD and XF Electric. The first trucks were delivered in August 2024 (DAF, 2024a). The DAF Electric is available only as a B-Truck. The XD and XF Electric is available as a B- and A-Truck (DAF, 2024b).

The trucks are equipped with LFP batteries, and 88% of the installed energy capacity can be used in operation. The DAF XB Electric can be equipped with 1-2 battery packs, resulting in installed battery capacities of 141-282 kWh. The DAF XD Electric can be configured with 2-5 battery packs, resulting in installed battery capacities of 210-525 kWh. The DAF XF Electric has 3-5 battery packs, resulting in installed capacities of 315-525 kWh. The battery packs are stored in battery boxes, which are attached with casted brackets to the chassis. The brackets are attached to the battery boxes on the left and right sides. Figure 6 shows the mounted bracket.



Figure 6: Casted Battery Bracket DAF

DAF has five different positions for the batteries, which are visible in Figure 7. Two battery packs each can be installed on the LHS or RHS of the chassis. On the LHS and RHS, they have two defined positions, which provide flexibility in fulfilling customer requirements. In the example of a truck with three battery packs, DAF can, e.g. install two battery packs on the LHS in positions 1 and 2. On the RHS, position 1 or 2 can be selected, depending on whether the customer wants free space to the front or back of the truck. Furthermore, they can place a battery pack below the driver's cab (DAF, 2024c).

As shown in Figure 7, the current electric BEV trucks of DAF are all equipped with an electric motor between the frame, which is connected with a propeller shaft to the rear axle. This setup is the same one Scania uses in their BEV trucks currently. The different electric motors provide a performance of 120-350 kW.



Figure 7: DAF XF Electric Packaging (DAF, 2025)

2.2.4 IVECO

IVECO launched their latest heavy-duty BEV truck, the S-eWay, in September 2024, using NMC propulsion batteries. The offering includes a B- and A-Truck.

The propulsion batteries of the B-Truck are stored in battery boxes in three different positions, see Figure 8. The battery boxes on the LHS and RHS are attached with brackets to the chassis. Furthermore, a smaller battery box between the frame can be installed. Depending on the configuration, 280-490 kWh nominal energy capacity will be stored in the battery boxes. On the LHS and RHS, two or three battery packs per battery box can be installed. The battery box between the chassis has one battery pack. This results in three different configurations of battery backs (LHS, in between, RHS), 2+1+2, 3+1+3 and 3+1+0 (IVECO, 2025).



Figure 8: IVECO S-eWay B-Truck Battery Layout (IVECO, 2025)

The truck is equipped with an E-Axle in the rear axle of the truck. The E-Axle contains two electric motors, which provide a standard performance of 480 kW and a peak performance of 840 kW (IVECO, 2025).

The S-eWay A-Truck of Iveco is equipped with the same E-Axle. The differences between the B- and A-Trucks are the amount and design of the battery packs. The A-Truck is equipped with 9 battery packs, which offer a total nominal capacity of 738 kWh. Three battery packs each are positioned on the LHS, RHS and in between the chassis. The battery packs of the A-truck are designed like cubes, which differs from the battery packs of the B-Trucks (IVECO UK, 2024).

2.2.5 MAN

MAN's latest electric truck models are the eTGS and eTGX can be configured as a B- or A-Truck.

The number of possible battery packs and the positions are the same between the models and can be seen in Figure 9. The trucks can be equipped with 4-6 NMC propulsion batteries, which result in a total nominal capacity of 320-480 kWh. If the truck has 6 battery packs, one battery pack (position 1) is positioned below the driver's cab. Furthermore, two battery packs (position 2 + 3) each are attached on the LHS and RHS between the front and rear axles. The battery packs in positions 1, 2 and 3 are the same. The position of the sixth battery pack (position 4) is different compared to the previous competitor analysis. The position of the drive unit, with a maximum performance of 400 kW, is compared to the competition, which does not use an electric axle, closer to the rear axle. This results in more space in front of the drive unit, which is why MAN positions a battery pack in front, between the frame. The drive unit is connected with a propeller shaft to the rear axle (MAN, 2025).

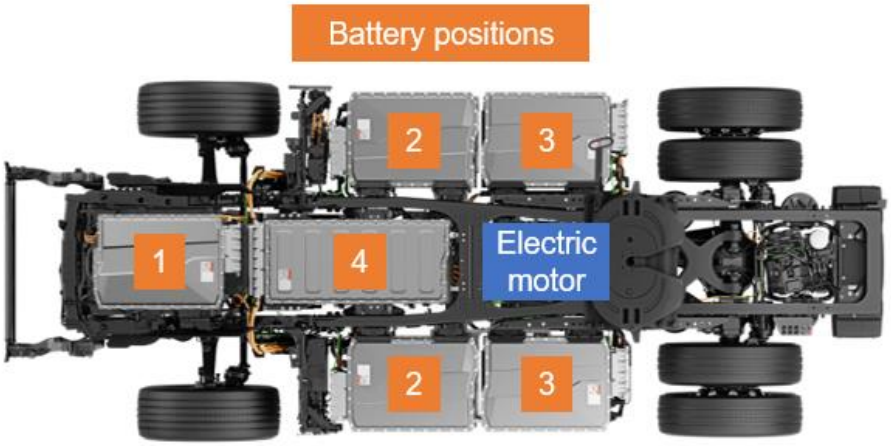


Figure 9: MAN eTGX Articulated Battery positions (Truckers World by MAN, 2024)

2.2.6 Competitor Comparison Summary

Based on the previous analyses of the BEV trucks of the competition and Scania, the analysis has shown that the battery suspension of Scania is unique on the market. All analysed competitors are suspending the propulsion batteries separately with brackets. The position of the battery packs of Scania is similar to most of the competitors, but none of them have a suspension connection between the LHS and RHS of the frame. As a result, the competition is more flexible in changing the position of the propulsion batteries or removing one. Furthermore, the competitors Mercedes-Benz and IVECO are using an E-Axle, which allows the installation of battery packs between the frame. Moreover, MAN achieves installing a battery pack between the frame, due to installing the powertrain further towards the rear axle. On top of that, MAN and DAF can offer a propulsion battery below the driver's cab, which Scania, in comparison, will not be able to offer before 2026 (see Section 2.1.2).

As a conclusion, the analysis has shown that single suspended propulsion batteries are possible and that a design such as the MS is not present in the market. Therefore, for the later conducted concept development, single suspended propulsion batteries can be considered in the idea generation.

Table 3: Competitor Comparison Summary

Attributes \ Competition		Scania	Mercedes-Benz	Volvo	DAF	IVECO	MAN
Truck Types	/	A- & B-Truck	A- & B-Truck	A- & B-Truck	A- & B-Truck	A- & B-Truck	A- & B-Truck
Battery Type	/	NMC	LFP	NCA	LFP	NMC	NMC
Nominal Battery Capacity	kWh	Up to 624	Up to 621 kWh	Up to 540	Up to 525	Up to 738	Up to 480
Battery position	/	LHS and RHS of the Frame	Underneath the Frame from LHS to RHS	LHS and RHS of the Frame	LHS and RHS of the Frame + under Driver's Cab	LHS and RHS of the Frame + between the Frame	LHS and RHS of the Frame + under Driver's Cab + between the Frame
Battery Suspension Type	/	MS	Battery Packs attached with Brackets	Battery Packs attached with Brackets	Battery Packs attached with Brackets	Battery Packs attached with Brackets	Battery Packs attached with Brackets
Number of battery packs	/	4	3	6	5	3	6
Powertrain	/	Powertrain + Propeller Shaft between the Frame	E-Axle	Powertrain + Propeller Shaft between the Frame	Powertrain + Propeller Shaft between the Frame	E-Axle	Powertrain + Propeller Shaft between the Frame
Powertrain performance	kW (continously)	450	400	490	350	480	400

2.3 Articulated and Basic Trucks

As mentioned above, Scania offers BEV trucks as A- and B-Trucks. The general design of the trucks is similar in many areas, but it differs in the following aspects, which is why different needs for the design result.

The fifth-wheel of A-Trucks is mounted on top of the chassis, at the end of the truck. The possibility to attach different trailers to it enables to transport of a wide range of goods. Regarding the goods, the payload of the truck is important, which defines how much weight the customer can load on the truck (Scania BIBAT, 2016d). The length of A-Trucks is important because it limits the possible maximum length of the trailer and thereby the space for goods. In the EU, the maximum allowed total length of A-Trucks with trailers is 18,75 meters (EUR-Lex, 1996). As mentioned earlier, in the development of the battery suspension for BEV trucks, Scania focused on accomplishing a space-efficient design to minimise the increase in truck length. In addition to the truck length, the range of A-Trucks is crucial, as they are primarily used for long-haulage applications. Therefore, the battery suspension must not only be compact but at the same time include a high battery capacity. This battery capacity aims to be able to drive with one charge 4,5 hours. Lastly, A-Trucks are ready to operate when leaving the production in Södertälje, because all required components to operate are mounted.

B-Trucks, in comparison, are not equipped with a fifth-wheel and are commonly not ready to operate when leaving the production in Södertälje. The reason for that is that B-Trucks require bodywork, which defines the application of the trucks. Thereby, the payload is crucial as well for certain applications. The bodywork is sometimes installed by Scania, but mostly by external bodybuilders. Due to the bodywork being installed on the truck's chassis, the truck itself is normally longer than an A-Truck. This is the case because the chassis of the truck needs to have almost the same length as the bodywork to handle the loads. Moreover, the bodywork is installed on top of the chassis, but in certain applications, free space on the side of the frame is required as well. Furthermore, the majority of B-Trucks are not used for long-haulage applications, but instead in urban applications. Therefore, shorter driving distances are commonly travelled per day, compared to A-Trucks (Scania Bodybuilder, 2023).

Due to the fact that the MS was developed to fulfil the needs of A-Trucks, the conclusion that the current design is sufficient can be made. The available battery capacities are capable of operating 4,5 hours without a break. Furthermore, the position of the MS does not collide with components on A-Trucks, because they require no additional bodywork. The MS is a good solution for long-haulage applications, but issues occur when B-Trucks are equipped with the MS. Therefore, the following investigation will focus on the needs for B-Trucks.

2.3.1 Bodybuilders

The bodybuilders are the responsible companies, who are installing the bodywork on the truck and thereby finalising the product. They are installing the primary load carrier and increasing the value of the truck, due to enabling it to operate in the customer's application. Depending on the application of the truck, the bodybuilders require certain preparations from Scania (Scania Bodybuilder, 2023). Figure 10 shows an example of the status of a B-Truck when arriving at a Bodybuilder.



Figure 10: B-Truck without Bodywork (Scania Bodybuilder, 2023)

When the bodyworks contains any movable or heated equipment, the bodybuilders need preparations on the truck to operate the equipment. Therefore, the PTO's (see 2.1.3) are installed on the truck to enable the operation of the bodywork. Typical applications for these are tippers, concrete mixers, refuse collectors, or trucks with an installed crane. Figure 11 shows a Tipper, which has the bodywork installed and is ready to operate.



Figure 11: B-Truck with Bodywork (Scania Bodybuilder, 2023)

For other applications of B-Trucks, the need to install components on the side of the frame often occurs. As a result, based on the application, the bodybuilders require free space on the side of the frame between the axles of the truck. Therefore, Scania needs to

prepare the truck so that the free space is available. Following, common bodywork will be investigated.

2.3.2 Bodywork

In the following, the examples for common bodyworks of B-Trucks, which require free space on the side of the frame, will be described, to understand where the free space is required. Furthermore, the problems which occur when offering the same applications with the current MS will be described.

Crane Applications

B-Trucks, which are equipped with a crane, require free space for side legs on the side of the chassis. Depending on the size of the crane and the weight it can handle, the dimensions and number of side legs vary. Commonly, one side leg each on the LHS and RHS behind the driver cab is attached to the frame, this installation requires a free space of approximately 600 mm. Before operating the crane, the side legs drive out to the side and then lift up the truck (Scania Bodybuilder, 2023). In the past, with ICE trucks, the truck was prepared in Södertälje, so that the required free space was available. Therefore, the components that are normally installed at the position on the side were moved further towards the rear axle or positioned at another location on the truck (Ytterstene, 2019). Due to the fact that the MS has a defined position, which would collide with the installation of side legs, Scania cannot offer BEV trucks with side legs installation.

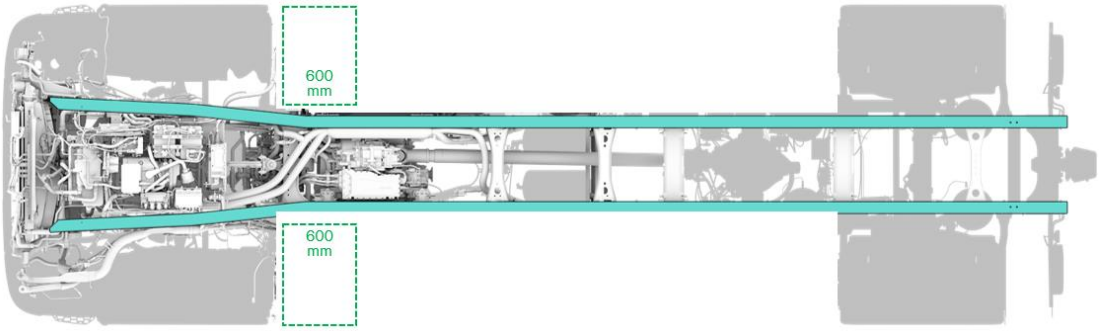


Figure 12: Crane Application Side Legs Installation

Refuse Collector (Side loader)

Refuse collectors can be equipped with a side loader, which pick up the waste bin from the kerb. These side loaders are installed on the RHS or LHS of the street, depending on whether the country has right or left-hand traffic. Besides the width of the side loader, the movement radius is important. The side loader moves first down, before then reaching forward and picking up the waste bin (Zöller-Kipper GmbH, 2025). Therefore, a free space of up to 1500 mm is required on one side of the frame for the installation (Ytterstene, 2019). For ICE trucks, this application was possible because the components on the chassis could be moved. With the BEV trucks, two problems occur. Besides that, the current position of the MS collides with the installation, an asymmetric MS design on the LHS and RHS is not possible due to the connection between the LHS and RHS of the MS (see 2.1.1). Therefore, when moving batteries backwards, the LHS and RHS need to be moved together. Furthermore, even when moving the MS backwards, so that it starts behind the side loader, the overall truck length would increase, which would result in other challenges. As a result, refuse collectors with side loaders are not applicable to the current design of the MS.

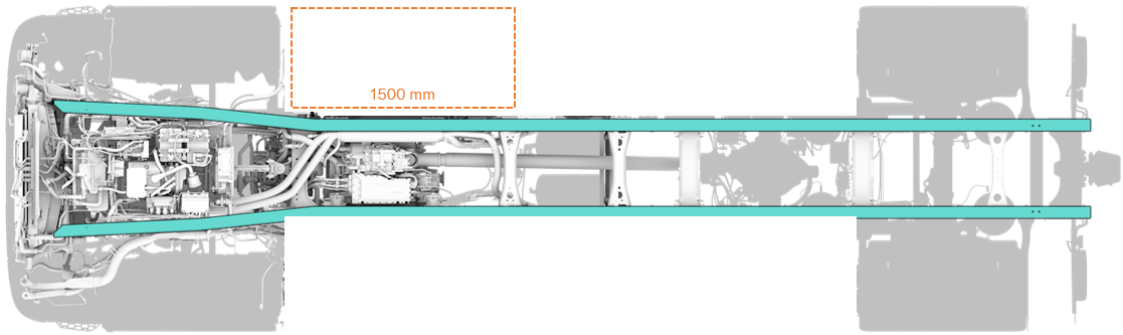


Figure 13: Refuse Collector with Side Loader (Right-hand traffic)

Road Sweepers

B-Trucks from Scania can be equipped with bodywork, which allows the application of Road Sweepers. The truck requires free space on LHS and RHS of the frame to be able to mount brushes, hovers and the required equipment to operate them. The equipment is attached to both sides of the chassis. Depending on the size of the road sweeper, the required space on the frame varies. Free space of 1500 mm to 2500 mm can be required on the frame of the truck (Ytterstene, 2019). The bodywork starts commonly behind the front axle, which is why it is not applicable with the current BEV trucks, due to the same reasons mentioned above.

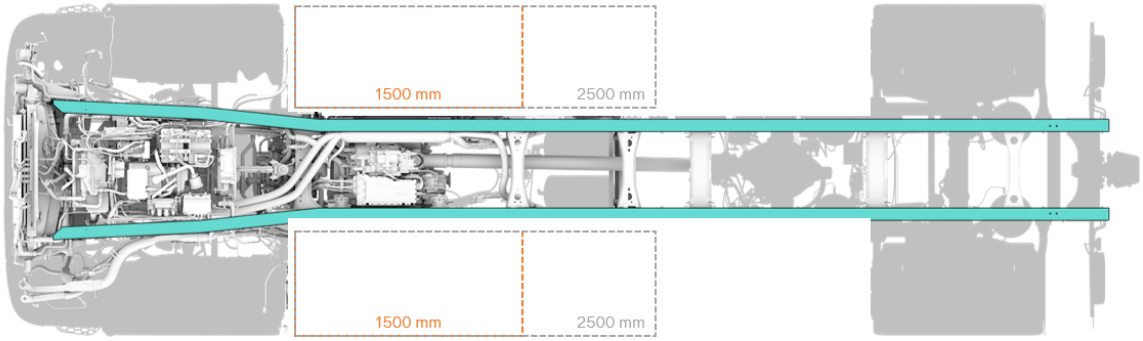


Figure 14: Road Sweepers

Material Box

A common bodywork on B-Trucks are material boxes. The number and position on the truck can vary. Often, the boxes are mounted on only one side of the frame, normally on the side where the driver is sitting. Due to this, the LHS of the frame requires a different layout compared to the RHS (Ytterstene, 2019). If the axle distance of the truck, in combination with the installed MS, offers enough space to install the material boxes after the MS, it can be offered with the current design. However, if the number of material boxes increases, an unsymmetrical design of the LHS and RHS is required, which cannot be realised with the MS. Moreover, the material box is often combined with other bodyworks, which is why the required free space on the frame for the material box is added.

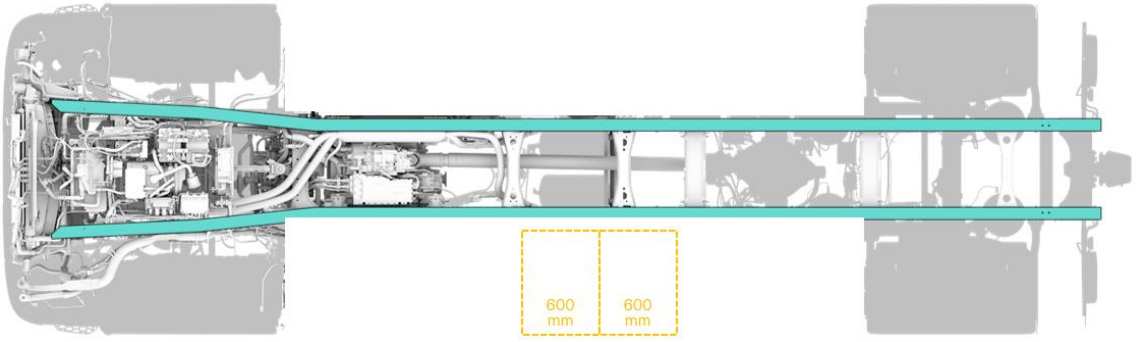


Figure 15: B-Truck with Material Boxes

Tanker

Trucks which are operating as tankers require specific free space on the side of the frame. The free space is required to fill and empty the truck with hoses. Therefore, a box is installed on the side of the frame, which contains hose connections. The attachment point can be between the front and rear axles or behind the rear axle. Depending on whether the country has right or left-hand traffic, the box is attached on the LHS or RHS of the truck. Based on the number of hose connections, the width of the box varies. Normally, a required free space on the frame of 2500 mm is considered when a B-Truck is designed for the application (Ytterstene, 2019). If the box is installed between the front and rear axles, it might collide with the MS, due to the limited free space on the frame. In certain configurations, an unsymmetrical design of the batteries might be required, so that more battery packs are on one side and fewer on the other side of the box.

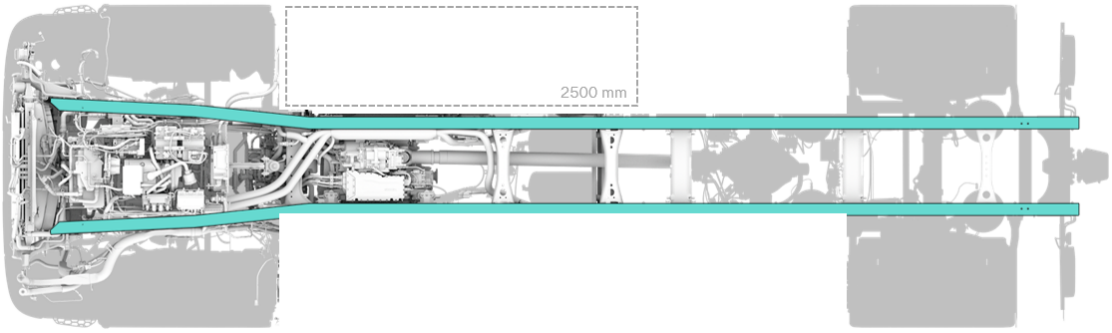


Figure 16: Tankers (Right-hand traffic)

Free Side

Lastly, the bodywork may require an empty side on the frame between the front and rear axles. If the RHS of the frame needs to be empty, all components which are normally mounted on the RHS were moved to the LHS or in between the frame, so that it doesn't collide with the bodywork (Ytterstene, 2019). When considering this situation with the MS, an empty frame side cannot be realised, because the MS is connected on the LHS and RHS of the frame. Removing one side is not possible because the design of the MS is based on being connected to both sides of the frame.

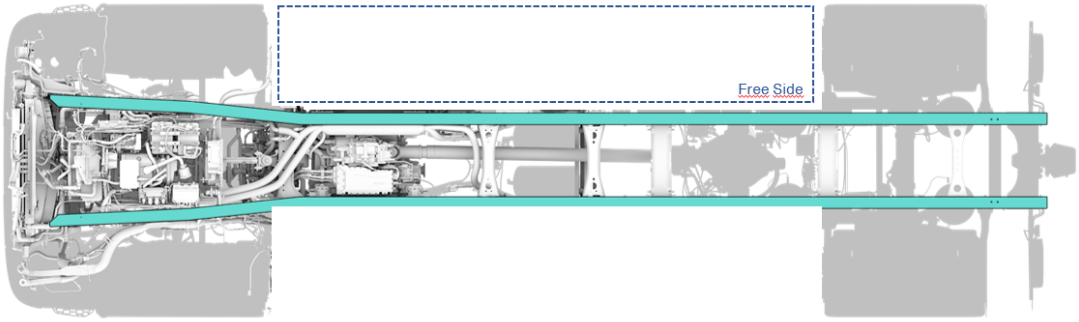


Figure 17: Free Side RHS

As a result of the different applications of B-Trucks, several positions for bodywork result, which are combined in Figure 18. It visualises the multiple positions of bodywork, which needs to be attached to the side of the frame, depending on the customer's application. Hence, Scania requires high flexibility when considering where to mount components to the side of the frame, to enable space for the bodywork. As mentioned previously, the MS has a fixed position and cannot be moved. Furthermore, the LHS and RHS of the MS need to be similar, which is why an unsymmetrical design is not possible. As a result, Scania cannot offer electric B-Trucks to the described applications, because the design of the MS doesn't enable the installation of the required bodywork.

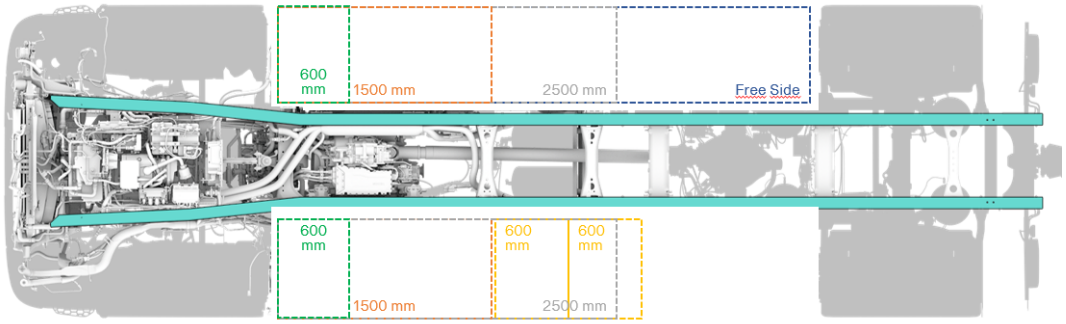


Figure 18: Combined Positions of the Bodywork

2.3.3 Space for Propulsion Batteries

Before developing a new battery suspension concept for B-Trucks, the available areas where to position the propulsion batteries need to be identified, because it will influence the design of the suspension.

As a start, the areas which are not available will be identified, see Figure 19. The length of B-Trucks is of importance, because it influences the manoeuvrability of the truck. Hence, B-Trucks are mainly operating in urban applications, and a shorter truck is beneficial. Moreover, the maximum truck width of 2,55 m must not be exceeded (EUR-Lex, 1996). As a result, the positioning of the propulsion batteries should not increase the truck length or width, which is why adding batteries at the front or at the back of the truck is not an option. Furthermore, because B-Trucks will be equipped with bodywork, the space above the frame is not available for propulsion batteries. When installing batteries above the highest point of the frame, it will collide with the bodywork or reduce the volume of the bodywork when the bodywork is built around the propulsion batteries. Therefore, it is a disadvantage to use the space above the frame.

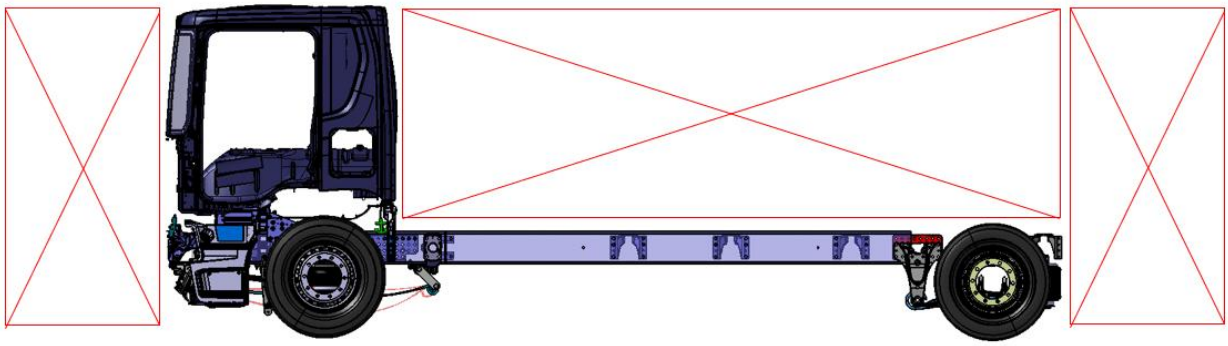


Figure 19: No space for Propulsion Batteries

When considering space where propulsion batteries can be installed, three major areas can be identified, see Figure 20. Firstly, underneath the driver's cab (1) is space for installing a battery (1). In the past, the space was used for installing the engine of ICE trucks. Since the powertrain is installed between the frame behind the driver's cab, the space beneath the cab remains available. Scania is currently working on a solution to fill the space with a propulsion battery (Kvaldén, 2025).

Secondly, when examining the truck layout, the space between the front and rear axles, within the frame (2), appears to be a potential area for installing propulsion batteries (2). A review of the competition (see Section 2.2, Market Analysis) reveals that some competitors, such as IVECO and MAN, are already occupying the mentioned space. Iveco is achieving it by using an E-Axle, which is installed between the rear axle. Therefore, the space between the frame is available to install a propulsion battery. MAN, in comparison, is achieving it by having a different frame layout and moving the powertrain further to the rear axle. As visible in Figure 9, the frame goes parallel at a later point, compared to

Scania. The frame stays wider behind the front axle to achieve the space for the propulsion battery. Behind the propulsion battery, the frame has a bend, which achieves parallel running beams of the frame. There, the powertrain is installed, which is connected with a propeller shaft to the rear axle. The discussed space is for Scania trucks not available, because the powertrain is mounted behind the driver's cab between the frame and is connected with a propeller shaft to the rear axle. Even though when moving the powertrain further backwards, the available space would be sufficient to install approximately 100 kWh of battery capacity. This capacity correlates with the installed capacity of a MAN truck (MAN, 2025). The additional design work which is required to move the powertrain backwards doesn't correlate with the gained advantages for this project. Furthermore, in addition to the extra design work, the area faces the highest competition for space, as components such as crossmembers, air tanks, and control units are also mounted there. In combination, Scania has no E-Axle, most of the space between the frame is not available (Stemp and Kvaldén, 2019). Therefore, this space will not be considered for the following ideation process.

Thirdly, the space on the side of the frame, between the front and rear axles, offers the biggest space to install propulsion batteries to the frame (3). As experienced above, the space on the side of the frame has big competition as well, when thinking of the bodywork of B-Trucks. Nevertheless, all the analysed applications which require bodywork have, after the installation of bodywork, free space for the installation of propulsion batteries (Kvaldén, 2025). Furthermore, due to the truck's length, there is the largest area to install propulsion batteries. Therefore, a battery suspension is required, which allows for different positions to install propulsion batteries on the side of the frame.

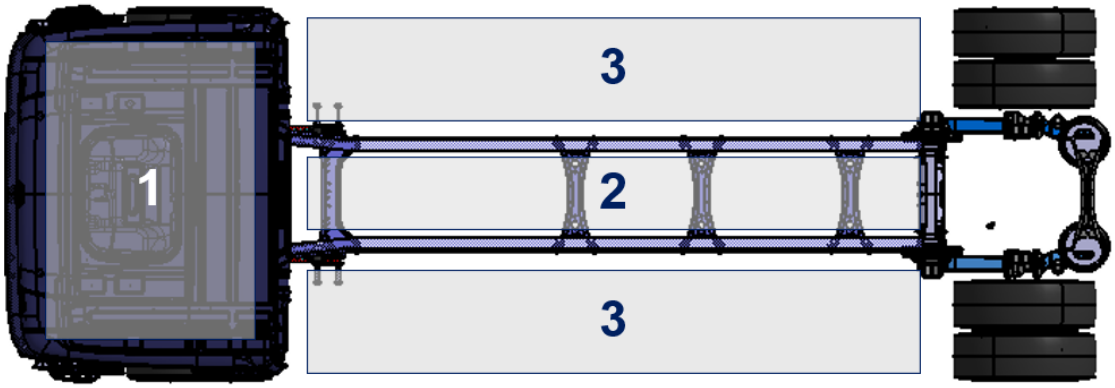


Figure 20: Space for the Installation of Propulsion Batteries

3. Methodology

The methods used in the concept development will be described in the following. The methods were used to guide the project team while developing a new concept of a battery suspension. Where the methods were applied and the reasons for choosing the methods will be clarified.

Quality Function Deployment

To translate the customer requirements into functional requirements, the QFD tool was applied. The QFD tool is known for guiding the design of a product towards the needs of the customers (Akao, K. Nagai and N. Maki, 1996). The approach of taking the needs of the customers and translating them into functional requirements is seen as an effective approach to achieve customer satisfaction (Matzler and Hinterhuber, 1998). Based on the insights gained from the Pre-Study regarding customer requirements, the QFD method is suitable for the concept development phase.

In the project, the customer requirements were defined by the previously conducted Pre-Study. The defined requirements were ranked by importance between 1 (low importance) and 5 (high importance). Based on these needs, the project team defined functional requirements, which fulfil the customer needs. To rank the priority of the functional requirements, the project team evaluated the relationship between the functional requirements and customer requirements, which means how strongly the functional requirements influence the successful fulfilment of the customer requirements. No relationship is evaluated with a 0, a weak relationship with a 1, a moderate relationship with a 3 and a strong relationship with a 9. The finished QFD tool can be seen in Appendix B.

Function Means Tree

For the generation of concepts, an FMT was developed. By applying the method, the project team gained a structural overview of what important aspects (Functions) need to be considered in the development of a battery suspension and how these functions can be fulfilled (Means). FMT are used to gain an overview of complex systems, because due to the structural approach, the main function gets divided into several functions, and by adding several solutions to the functions (Means), the number of ideas increases (Bracewell, 2002). To consider a high number of possible concepts, applying the FMT method supported the project team, which is why the method was applied. The FMT can be seen in Appendix C.

Based on the finished FMT, the project team created combinations of possible concepts. Therefore, one mean per function was combined with the other, and all possible

combinations were generated, see Table 5. Following, the combinations were evaluated by the project team to determine if they are feasible or not. Based on the judgment, the feasible combinations were used as an inspiration when developing drawings of the design of the battery suspension.

Pugh Matrix

To evaluate the generated concepts, the Pugh Matrix was used. The tool is used in research for evaluating concepts, based on criteria which were defined before (Pugh, 1991). Therefore, the customer requirements of the QFD tool were used to evaluate the concepts. One concept was set as a standard, to which the other concepts were compared. Based on whether the concept fulfils the requirements worst (-), the same (0) or better (+), the concept gets evaluated. The concept with the highest total points, by which the evaluation worst gives minus points, the same no points and better gives plus points, was the chosen concept which was developed further. The finished Pugh Matrix can be seen in Table 6.

Catia V5

The chosen concept were developed in Catia V5, which stands for Computer-Aided Three-dimensional Interactive Application (Dassault Systems, 2025). Catia V5 is used at Scania for the product design and development. Besides the design of the battery suspension in Catia V5, the software was also used to analyse the battery suspension. Therefore, a Generative Assembly Structural Analysis was conducted to optimise and confirm the design of the concept. During the analysis, simulations were applied, which showed the stress (MPa) in the components. Based on these results and in consideration of target values, the design can be evaluated.

4. Requirements

Based on the findings of the Pre-Study, the requirements for the design of a new battery suspension will be defined in the following. The Pre-Study showed that the design of a new battery suspension should concentrate on the needs of B-Trucks. Therefore, a QFD will be conducted to identify, based on the customer requirements, the functional requirements for developing a new concept of a battery suspension.

4.1 Requirement Specification

The QFD tool was used to translate customer requirements into technical specifications and design parameters, see Appendix B.

The diagram helps to visualise the relationship between the customer requirements and how the development can fulfil them. The customer requirements and functional requirements will be described in the following.

4.1.1 Customer Requirements (What's)

To start applying the QFD method diagram the customer requirements have to be identified. Based on the findings of the Pre-Study, 9 different customer requirements were identified, shown on the left side of the diagram.

Maximising Payload

As mentioned in the Section 2.3, payload refers to the weight customers can load onto the truck without exceeding the maximum axle load. The maximum payload is limited by the total weight of the chassis. The lighter the truck, the more payload it can carry. Even the ability to load just a few kilos more per trip can lead to significant gains across an entire fleet. Therefore, when developing a battery suspension, the weight of the total installation should be optimised towards weight efficiency.

Different positions of the Battery Suspension

As learned in the Section 2.3.2, B-Truck customers require the option to choose different positions for the battery suspension. This flexibility is important because trucks are used in many different ways, each with its own set of needs. Offering various positions where the suspension can be mounted can not only help increase the truck's payload but also make it easier to install custom bodywork, allowing the truck to be tailored to the specific requirements of the job. This means the design of a new battery suspension must allow for flexible mounting positions along the frame.

Different ranges (km)

Customer's daily operations of B-Trucks can look very different, some drive long distances every day, while others cover much shorter routes. Nevertheless, customers of B-Trucks typically drive fewer kilometres per day on average compared to A-Truck customers. To support these varying needs, it's important to offer different battery capacities to suit the different required ranges per day.

Robust

A robust design is important for customers because it ensures that the truck can withstand the demands of different environments and operating conditions. Trucks in general are exposed to heavy workloads, so having a durable design helps prevent damage and reduces the need for frequent repairs. This also lowers costs in the long run.

Safe

The customers require a battery suspension which is safe to operate. This means that it has to be mounted safely to the truck, without the risk of hurting operators or other passengers on the road.

Affordable

When it comes to customers, affordability is another important factor when investing in a truck. This is because many customers rely on cost-effective solutions, which is why the solution of a new battery suspension should be cost-effective.

Enabling Bodybuilder adaptations

Bodybuilders are one of the key customer groups, so it's important to ensure they have enough free space to carry out their specific bodywork installations. Their work often involves adding the main load carrier and making other adjustments to match the truck to the customer's actual needs. Making it easier to adapt and integrate the bodywork helps ensure the truck is ready for real-world use.

Free space on the frame

As mentioned above, having free space on the frame is important for meeting different customer applications. Where the free space is required varies, which is why flexibility is required, to change where the free space on the frame is.

Legal to operate

A new battery suspension needs to be legal to operate, which is why a new design needs to follow the legal requirements.

4.1.2 Functional Requirements (How's)

The functional requirements (How's) on the top part of the diagram represent the translation of the customer requirements into technical requirements. The functional requirements will be presented next. Thereby, explanations will be given for how the functional requirements fulfil the customer requirements. The ones which were evaluated with a strong relationship of 9 on the QFD will therefore be explained.

Minimum required reinforcement

To have a minimum required reinforcement ensures that the design can support the expected loads and stresses without being over-engineered. At the same a minimum reinforcement solution is necessary to reduce costs and at the same time maximise payload, because of the lower weight of the battery suspension.

Lowering the stress on the frame

Lowering the stress on the frame is also important to consider for several reasons. When the stress is reduced, the frame is less likely to experience fatigue, cracks, or permanent deformation. This not only improves safety and robustness but also reduces the risk of costly repairs.

Flexible attachment points on the frame

Having multiple options for where the batteries should be attached to the frame is important because it gives a lot of design flexibility. Every truck and its purpose can be a little different, so being able to choose where to place the batteries makes it easier to customise the design based on the truck's specific needs, whether it's about saving space, improving weight distribution, or fitting other components. Furthermore, the special order department requires the flexibility to be able to address the different customer needs.

Up to 300 kWh battery capacity

Considering that B-trucks operate mostly in urban areas, they aren't typically driven over long distances. Instead, they operate in cities, handling local tasks where the driving range is limited. Because of this, they don't require as much battery capacity as long-haul trucks. One important aspect of the design is therefore to ensure that the suspension system can support a battery capacity up to 300 kWh. This is nearly half of the capacity of the MS, and when considering that the trucks operate in urban applications, this capacity is considered enough to meet the daily energy demands of urban operations while also

allowing room for different needs depending on how the truck will be used. At the same time, the solution will stay affordable because the propulsion batteries are leading to higher costs, which is why a lower capacity is beneficial. Moreover, fewer propulsion batteries lead to a lower weight of the installation, which is positive for maximising the payload.

Unsymmetrical battery design

It is also important to have an unsymmetrical battery design to be able to meet different customer requirements. By placing the batteries asymmetrically on LHS and RHS of the frame, there will be free space on the opposite side where no batteries are placed to install. This space can then be used for bodywork, providing more flexibility in the truck´s design.

Compact battery suspension design

Another requirement is to have a compact battery suspension design because it benefits the space efficiency. By having a more compact design, less room is needed for the battery suspension, which allows more space for other components and generates more free space.

Safe storage of the batteries on the truck

The batteries should also have safe storage on the truck, for example, using side protection to keep the batteries secure, which helps prevent accidents. Batteries can be dangerous if they’re not properly stored, and if they’re not secured well enough.

Chassis width maximum 2550 mm

The chassis width must not exceed 2550 mm, and the design must stay within this limit. Going beyond this would exceed the legal requirements in Europe, which is why the truck would not be legal to operate.

Several attachment points for the batteries

It’s important to have several attachment points on the frame for the batteries, as this allows for a more flexible and adaptable design. With multiple mounting options, the batteries aren’t restricted to just one fixed location, which means they can be placed in a way that best suits the overall layout of the truck. This flexibility not only makes it easier to optimise weight distribution and space usage, but it also opens up room for other essential components to be integrated more efficiently.

Gross vehicle weight (GVW)

GVW is the total weight of the vehicle, including both the vehicle itself and any cargo it’s carrying, just not including any trailers attached. The new design of the battery suspension should concentrate on achieving a low weight to maximise the payload for the customers and to fulfil the legal requirements.

4.3 Project Targets

The project aims to develop a new battery suspension concept for Scania's electric B-Trucks. Based on the learnings of the Section 2.3.3, the concept will focus on mounting the battery suspension to the side of the frame, between the front and rear axles. Thereby, the focus is to enable the different bodywork adaptations which were described above (see Section 2.3.2). To accomplish this, the design will be guided by the QFD tool, with a special focus on achieving a design which enables an unsymmetrical battery installation, with the flexibility to be mounted at different positions on the truck, as they were ranked as the top two priorities in the QFD, see Appendix B. The concept development will focus on finding a battery suspension concept for the MP10 battery. This reduces the complexity, and at the same time, the weight of the MP10 is lower (see Section 2.1.2), which makes it more reasonable that the frame can handle the stress. The described concept development aims to achieve a solution for the S-Order department within Scania, to solve the requests of their customers. Therefore, the developed concept should consider that it is possible to be implemented without the need for expensive and time-consuming development work. By achieving that, the chance that the developed concept will be further developed and implemented in the future increases.

5. Implementation

This chapter explains the design generation and concept evaluation process, where creativity plays a key role in generating as many ideas as possible. Due to the complexity of designing a complete battery suspension, the project team decided, as a first step, to conduct a Pre-Concept Development Work. Thereby, the focus is to find concepts, how to grab the battery and attach it to the frame. After developing ideas, the ideas will be presented to a design Engineer within Scania, who will be able to give feedback in terms of feasibility. Based on this feedback, the project team concentrated on one solution, how to grab the battery and developed several ideas with the support of the FMT method. The resulting concepts were evaluated by the Pugh Matrix to find the concept which suits the customer requirements of the QFD tool the best. The winning concept was taken and developed further.

5.1 Pre-Concept Development Work

This phase started with an open brainstorming on where to hold the battery. Thereby, the project team only focused on the mentioned and didn't consider any limitations which may occur because of the complexity of a battery suspension. By that, the variety and creativity of the ideas were intended to be increased. To simplify the description of the ideas, the surfaces of the batteries are defined by letters, as in Figure 21.

Side	Explanation
A	Back side of the battery, facing the frame
B	Top of the battery
C	Front side of the battery, outside of the truck
D	Bottom side of the battery, facing the road
E	Right side of the battery, facing rear axle
F	Left side of the battery, facing front axle

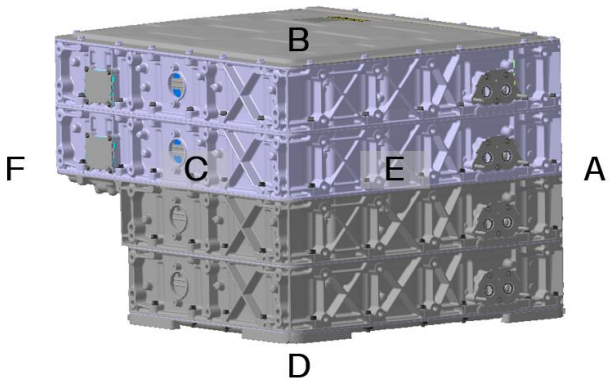


Figure 21: Alphabetical Numbering of Battery Sides

5.1.1 Around the Battery

Two ideas concentrate on holding the battery by having an attachment which goes around the battery.

Idea 1 is to hold the battery by having two attachment stripes starting from A, going over B, C and D and ending at A again. To handle the weight of the battery and reduce the stress in the frame, two attachment stripes were considered in the idea (see Figure 22). This idea is similar to how fuel tanks are grabbed when getting installed on ICE trucks.

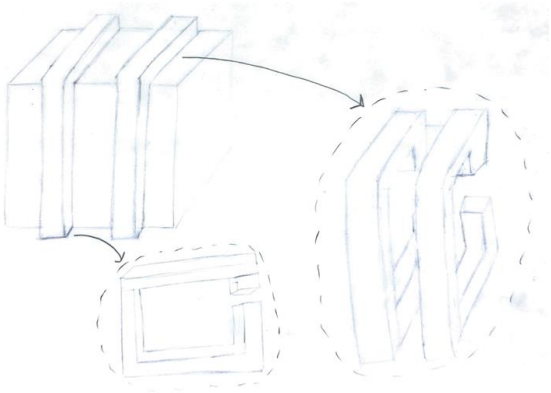


Figure 22: Around the Battery – Idea 1

Idea 2 is based on the same principle, but in this case, the battery is wrapped horizontally over the sides of the battery (see Figure 23). Therefore, the attachment starts at A, going over F, C, E and ending at A again.

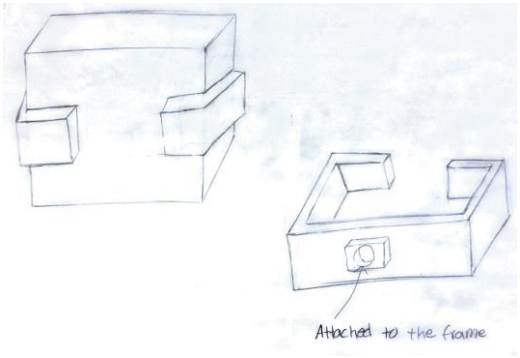


Figure 23: Around the Battery – Idea 2

Idea 3 has similarities to idea 1, but instead of two attachment stripes, a housing for the battery was designed. The housing is on the same sides as in idea one, and on top of that, a design of how to attach the housing to the frame was ideated. The idea can be seen in Figure 24.

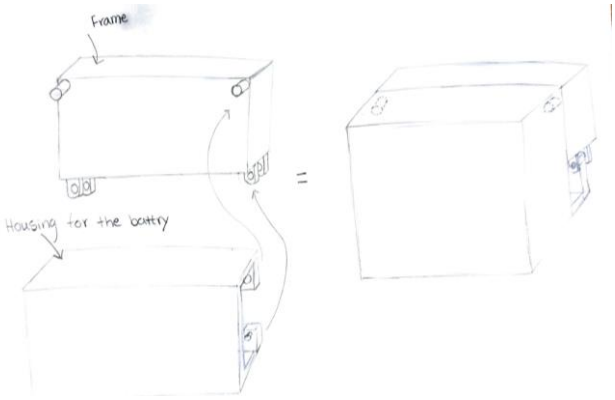


Figure 24: Around the Battery – Idea 3

5.1.2 Back of the Battery

Several ideas were generated, where the battery is held only on the back side of the battery (A).

Idea 1 holds the battery at A with an attachment, which has the width of the battery and the height of the frame, to mount it to the frame (see Figure 25).

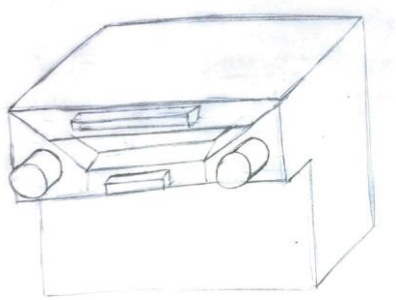


Figure 25: Back of the Battery – Idea 1

Idea 2 has three attachments at A, which have the same height as the battery. The idea considers the connection to the frame as well by hanging the battery on the frame. This attachment was inspired by how the forks of a forklift truck are attached to the fork carriage (see Figure 26).

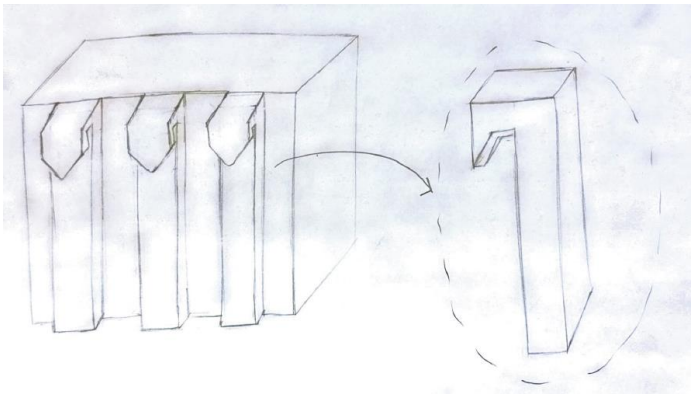


Figure 26: Back of the Battery – Idea 2

Idea 3 has similarities with idea two, but the three attachments wrap furthermore around a bit at side B and D. Furthermore, the attachments are screwed to the frame, see Figure 27.

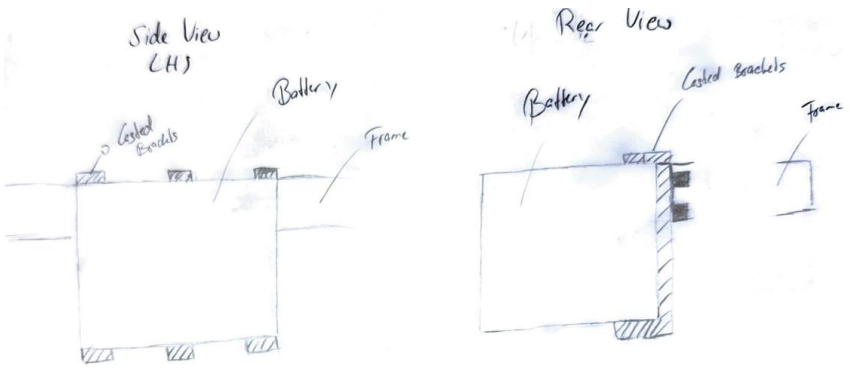


Figure 27: Back of the Battery – Idea 3

5.1.3 Side of the Battery

Three different ideas were developed, which hold the batteries on the left and right side of the battery (E + F)

Idea 1 uses the bulkheads of the MS, which are attached to sides E and F of the battery. To the top corner of the bulkheads on sides E + F is an attachment, which is connected behind the battery on side A (see Figure 28).

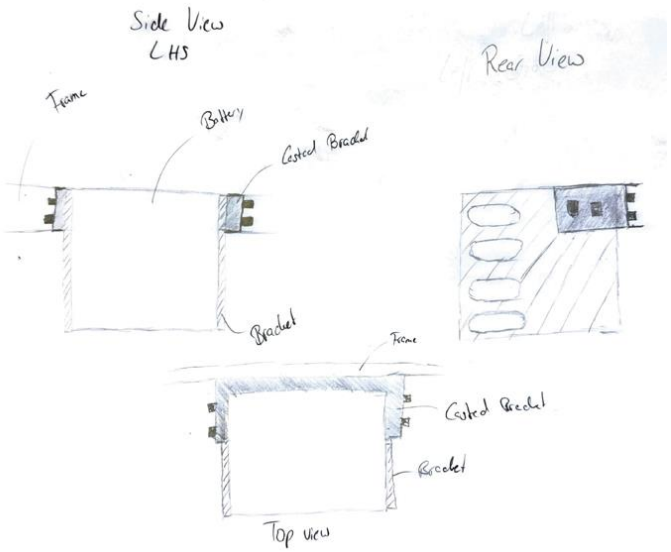


Figure 28: Side of the Battery – Idea 1

Idea 2 uses the bulkheads in the same position, but a beam is connected to both sides, which goes through the first side member of the frame and ends inside the second side member (see Figure 29).

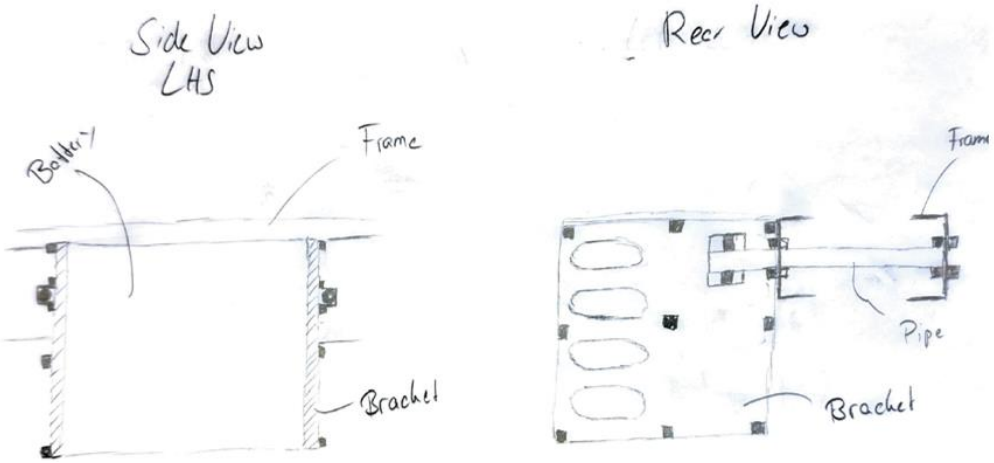


Figure 29: Side of the Battery – Idea 2

Idea 3 is attached with two beams, at the top and bottom, each on sides E + F of the battery. The top beam is connected to the side member of the frame, and the second beam is connected with the help of a bracket to the frame. The bracket is attached to the frame and continues down to the level of the second beam (see Figure 30).

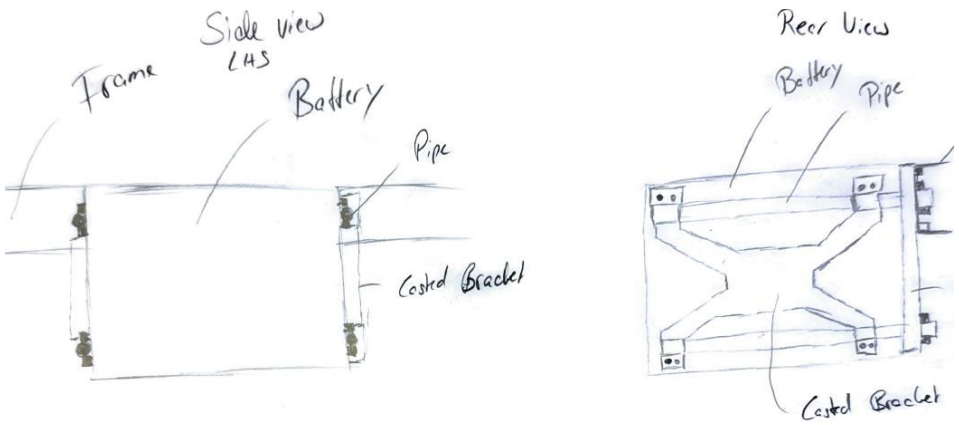


Figure 30: Side of the Battery – Idea 3

5.1.4 Pre-Concept Development Learnings

During the Pre-Concept Development Work, the project team gained valuable insights. By ideating different ideas on how to grab the propulsion battery, the project team received a better understanding of the batteries and the frame. By developing different ideas of how a connection to the battery and the frame can be achieved, different ideas were generated, which will influence the further development.

Moreover, the ideas were presented to a Design Engineer within Scania, who has experience with the MS (Borreani, 2025a). With this expertise, the project team was able to receive valuable insights, which defined how to continue. During the presentation of the ideas, the project team learned the following aspects:

- When suspending a component to the frame, all the connections together should result in a closed loop. By achieving this, the stress within the frame is better spread, and the chance that the frame can handle the stress increases.
- The MS has, due to the propulsion batteries, a high stiffness.
- The propulsion batteries have interfaces on the side of the battery (E + F, see Section 5.1) to be connected to a bracket. In the MS, the bulkheads are connected to these sides.
- A lot of development time went into the design of the bulkheads so that they can handle the stress when suspending the propulsion batteries to the bulkheads.
- The bulkheads have two connection points
    1. Connection at the top to be mounted to the frame by a clamping bracket and beam
    2. Connection at the bottom, which is connected to the other side of the MS, to prevent movement of the batteries below the frame
- When suspending the battery on only one side of the frame, the resulting stress on the frame may exceed the allowed maximum values. Therefore, how to reinforce the frame is important to consider.
- Scania uses maximum stress values (MPa) which are not allowed to be exceeded in stress simulations

With these findings, the project team decided not to consider further the presented ideas in 4.1.1 and 4.1.2. Instead, the concept generation will concentrate on developing concepts, which hold the MP10 on the sides (E + F). Therefore, the bulkheads will be used as a connection to the battery, because a lot of development went into the design, and it is validated that the parts handle the stresses. Due to the fact that the bulkhead has an interface for a clamping bracket, the part will be used as well. Lastly, the clamping bracket clamps a beam in the MS, the project team will consider beams as well. This decision was made to be able to achieve a high amount of similar parts between the MS and the new design, to lower the cost of the new battery suspension. This supports achieving the mentioned goal in the Section 4.2, that the solution will be able to be implemented within

the S-Order department. Following the bulkhead, clamping bracket, and beam will be described in detail, to identify the interfaces and attributes of the designed parts, to be able to generate concepts afterwards.

5.1.5 Bulkhead

Before the concept generation, the bulkhead needs to be analysed to identify the interfaces. On an MP10 battery, two bulkheads are installed, which are mounted on the left and right sides of the battery. The interfaces (1 and 2) are at the same positions on both bulkheads, only the cut-outs at the outside (3) differ. Figure 31 shows the bulkhead, which is mounted at side F of the battery, see 5.1.

At the top right corner of the bulkhead (1), the interface for the connection to the beam is located. At the bottom of the bulkhead (2) is an interface located, which makes the bulkhead wider at the bottom, compared to the top. This interface is used to mount a beam to it, which is connected in the MS to the bulkhead on the other side of the frame, to prevent the movement of the batteries under the frame. Because of the weight of the battery, without that connection, the battery would move, which would result in a lot of stress in the frame, see Section 2.1.1. Furthermore, there are several cut-outs in the bulkhead (3), which are used for the VCB cables or cooling hoses.

With the mentioned design in mind, the concept development needs to consider that the bulkhead needs two connections (at 1 & 2) and half of the bulkhead (3) cannot be used for any attachment. Furthermore, when suspending an MP10 battery, two bulkheads in total are needed, which are attached to the mentioned sides of the propulsion battery.

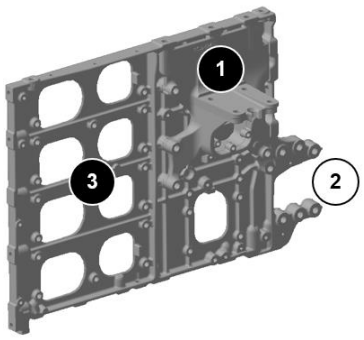


Figure 31: Bulkhead

5.1.6 Clamping Bracket

Due to the usage of the bulkhead, with the interface at position 1 to be connected to a beam, the clamping bracket of the MS will be used as the interface to connect the beam with the bulkhead. The clamping bracket is connected, besides to the bulkhead, to the frame of the truck. The clamping bracket consists of several layers, with bearings included, which is why the clamping bracket suspense the battery suspension. Two clamping brackets in total will be needed, to be able to attach one clamping bracket each to the two bulkheads. Due to the development work, which went into the design and the fact that the clamping bracket successfully suspends the MS, the project team decided to use the clamping bracket as well. Therefore, the complexity gets reduced, and at the same time, another similar part, compared to the MS, gets added to the new design of the battery suspension. Figure 32 shows the clamping bracket (marked in orange) connected to the bulkhead and the frame.

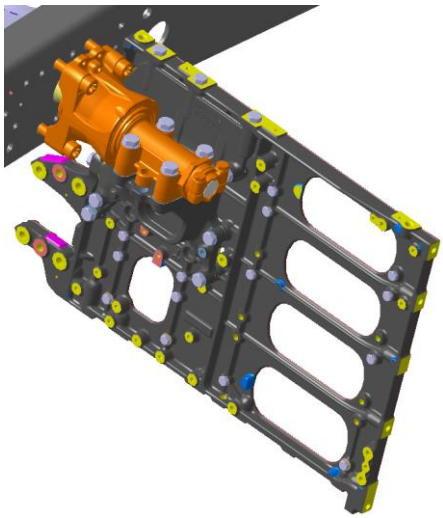


Figure 32: Clamping Bracket mounted to the Bulkhead and the Frame

5.1.7 Beam

As a result of using the bulkhead and clamping bracket for the development of the battery suspension, the project team decided to use a beam as a connection to the frame. This decision was made because the clamping bracket has an interface, which is prepared for the connection to a beam. The beam can be seen in Figure 33, marked in orange. For the following concept development, the length of the beam is not defined, only the usage of a beam is a prerequisite for the concept development. Furthermore, two beams are required to suspend one MP10.

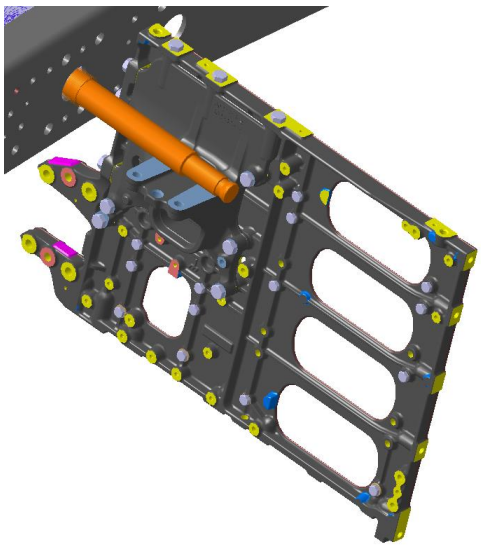


Figure 33: The Beam located in the clamping bracket

5.2 Concept Generation

To start with the concept generation, the FMT method was applied to gain a better understanding of how to design the battery suspension (see Appendix C). Applying the methodology helped the project team to consider a variety of different approaches to achieve a solution for the identified problem.

The main function (F1a) is to develop a flexible battery suspension, and the position has to be on the frame, between the front and rear axle, which is why this is the first mean (M1a). This was identified during the analysis in the Section 2.3.3.

Based on the learnings of the Section 5.1.4, the first three functions, F1b1, F1b2 and F1b3, have only one mean. Because the bulkhead is used as the connection to the battery, the clamping bracket is used, and in combination, a beam is required. This is due to the MS design, which uses the mentioned three components. Furthermore, using the mentioned three components will reduce the development costs of the battery suspension and will make the new design more cost-effective. Therefore, M1b1.a1, M1b2.a1 and M1b3.a1 results as the only means.

The third function (F1b3) is also a key learning from the Section 5.1.4. Reinforcing the frame to withstand the stresses introduced by the battery suspension was identified as one of the main challenges. Therefore, during brainstorming, different means were identified, how a reinforcement of the frame can be achieved. The described functions and means can be seen in Table 4.

Table 4: Function and Means Concept Generation

Functions		Means	
F1b1	Connection to the Frame	M1b1.a1	Beam
F1b2	Connection ot the Battery	M1b2.a1	Bulkhead
F1b3	Connection to the Battery Connection	M1b3.a1	Clamping Bracket
F1b4	Frame Reinforcement	M1b4.a1	Bodybuilder Frame
		M1b4.a2	Double Frame
		M1b4.a3	Crossmember
		M1b4.a4	Connection to the other side of the Frame
		M1b4.a5	No Reinforcement

Based on Table 4, the project team started to brainstorm how to design the battery suspensions. Therefore, drawings were created, which included the mentioned means from above. All possible combinations with the means of Table 4 were created at least

once, and some of the combinations were used multiple times, because the project team created several drawings for the combination. As a result, the project team created a total of 8 different concepts, which are summarised in Table 5.

Table 5: Function Means Tree Combinations

Combinations				Concept
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a1	1
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a2	2
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a3	3
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a3	4
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a4	5
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a4	6
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a5	7
M1b1.a1	M1b2.a1	M1b3.a1	M1b4.a5	8

In the following, the different concepts will be described. All the figures are designed from the perspective of the rear axle, looking to the front axle. Nevertheless, the visualised suspensions are attached on both sides of the battery to the bulkhead, but the following figures only visualise the right side of the battery, which is equivalent to side E of the Section 5.1.

5.2.1 Concept 1

Concept 1 uses the bodybuilder frame as an additional connection point for the suspension of the battery. Bodybuilders are mounting an extra frame (5) on the frame of Scania (4), to be able to mount the bodywork. The upper beam (2) is connected to the bulkhead (1) and the frame of the bodybuilders (5). As a result, the lower beam (3), which is connected to the bottom of the bulkhead, can be attached to the frame of the truck (4). This setup gives the bulkhead solid support from both above and below, making the whole structure more stable, see Figure 34.



Figure 34: Concept 1

5.2.2 Concept 2

Concept 2 is based on the idea of strengthening the truck's frame by adding an extra layer, making it thicker and more robust. The so-called double frame (3) can carry heavier loads without the risk of breaking. In this concept, the bulkhead (1) is mounted at the top using a beam (2) which is attached to the inside of the left frame member. Furthermore, to prevent the movement of the battery below the frame, a U-shape connection from the LHS to the RHS of the frame is designed. On the LHS, it is connected to the bottom part of the bulkhead (5) and continues from there horizontally to the RHS, not to collide with other parts, which are located between the frame. Figure 35 shows Concept 2.

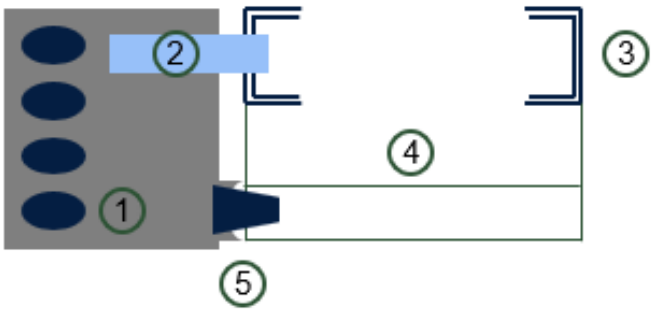


Figure 35: Concept 2

5.2.3 Concept 3

Concept 3 considers the mounting of one battery each on the LHS and RHS of the frame, and tries to find a modular design. The batteries are mounted as in concept 2 with the bulkhead (1) and a beam (2) attached at the top. The beam continues through the frame and is secured from the inside. Moreover, the same U-shape connection (4) between the LHS and RHS of the frame exists, to which the bottom part of the bulkheads (5) are connected. Moreover, the concept has an additional crossmember (3) between the frame to achieve a higher strength of the frame, to handle the resulting stresses of the battery suspension. The concept considers a modular approach because the goal of the design is to make it possible to mount just one battery on one side of the frame, without needing support or reinforcement from the opposite side. At the same time, the design allows for a second battery to be added on the other side if needed, which can be seen in Figure 36, without requiring any major changes, making it a solution that works well for both single- and dual-battery configurations. This is achieved due to the U-shape connection (4) and the interfaces at the bottom, which can be connected to the bulkhead at the bottom (5).

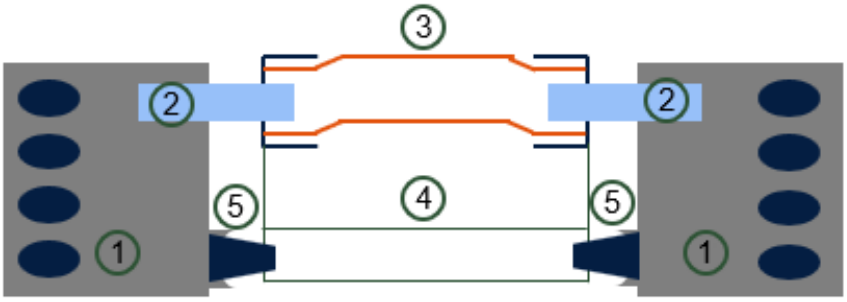


Figure 36: Concept 3

5.2.4 Concept 4

Concept 4 uses a U-shaped crossmember (3), which enables the connection to the bulkhead at the bottom (4) and at the same time reinforces the frame. Moreover, the U-shape was designed so that the propeller shaft runs above it and doesn't collide. Due to the connection of the crossmember to the RHS of the frame, the load of the battery suspensions aims to be transferred to reduce the stress on the LHS of the frame. The connection of the beam (2) at the top of the bulkhead (2) is the same as in the previous concepts. Figure 37 visualises the described concept.

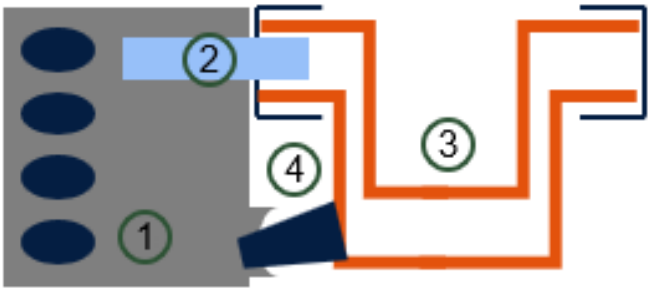


Figure 37: Concept 4

5.2.5 Concept 5

Concept 5 has attributes which has been seen in the previous concepts, see Figure 38. The U-shape connection (4) between the LHS and RHS of the frame enables the connection to the bulkhead at the bottom (5). The difference occurs in the design of the beam and the connection points. The beam is connected at the top of the bulkhead (2), but instead of being connected at the inside of the LHS, it continues to the inside of the frame member on the RHS (3). This design aims to transfer the load of the battery suspension to the RHS of the frame, to lower the stress in the LHS.

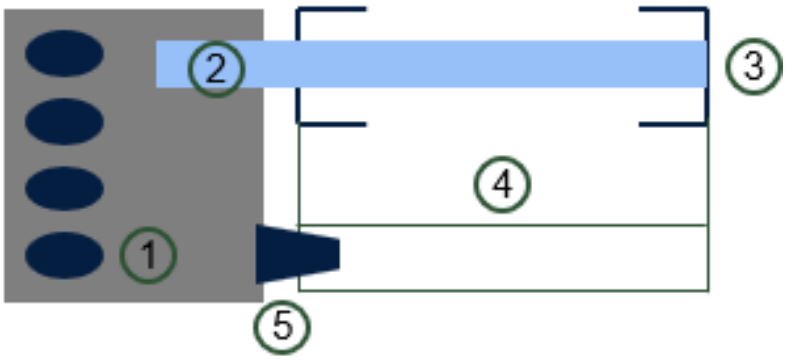


Figure 38: Concept 5

5.2.6 Concept 6

Concept 6 has a lot of similarities with Concept 2. The difference occurs when concentrating on the frame, see Figure 39. Concept 6 has the standard frame and does not use the double frame, as Concept 2.

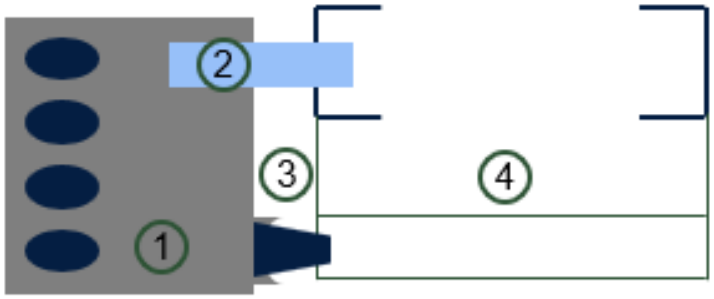


Figure 39: Concept 6

5.2.7 Concept 7

This concept is designed with three attachment points, all concentrate on one side of the frame, see Figure 40. The first two points are on the bulkhead (1), one at the top (2) and one at the bottom (4). The third attachment uses a bracket (3) that starts from the top inside of the frame, where the beam (2) is connected and continues down to the bottom of the bulkhead (4). The bracket aims to take the load of the movement of the battery when trying to twist under the frame.

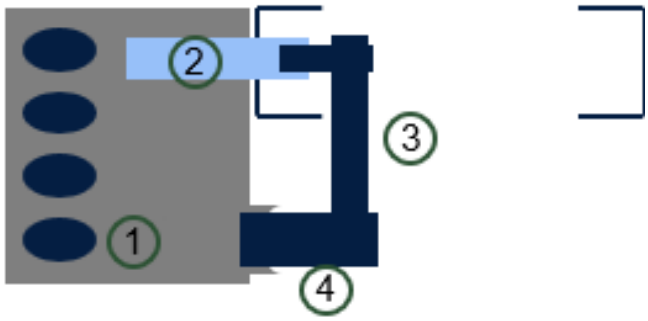


Figure 40: Concept 7

5.2.8 Concept 8

Concept 8 builds on the same attachment method used in Concept 7, where the bulkhead is connected to the frame at the top and bottom. The key difference here is an additional beam (5) that extends from the lower part of the bulkhead and runs underneath the frame. This beam is then secured to the other bulkhead, adding extra support and stability to the overall structure.



Figure 41: Concept 8

5.3 Concept Evaluation

A Pugh Matrix was used to evaluate the different concepts based on the requirements from the QFD. Each concept was compared to a reference concept to assess how well it could meet these requirements. The evaluation was done using symbols: + indicated better performance, – represented worse, and S showed similar performance. The comparison showed the strengths and weaknesses of each concept, making it easy to see which ones worked well in important areas, which needed some improvement, and which one was the best to move forward with, see Table 6.

Table 6: Pugh Matrix Evaluation

Criteria		Concept							
		1	2	3	4	5	6	7	8
1	Maximising payload	Reference	-	-	-	-	-	-	-
2	Different positions of the Battery Suspension		S	-	-	-	+	+	S
3	Different ranges (km)		S	S	S	S	S	S	S
4	Robust		S	+	+	+	+	+	S
5	Safe		S	S	S	S	S	S	S
6	Affordable		-	S	S	S	S	+	S
7	Enabling Bodybuilder adaptions		+	+	+	+	+	+	+
8	Free space on the frame		S	S	S	S	S	S	+
9	Legal to operate		S	S	S	S	S	S	S
Total -			2	2	2	2	1	1	1
Total +			1	2	2	2	3	4	2
Result			**-1**	**0**	**0**	**0**	**2**	**3**	**1**

To evaluate how well the various concepts met the requirements, the different concepts were presented and discussed with colleagues, with sketches shown to gather feedback and guidance. Valuable input was collected during these discussions to further refine the concepts, which are described in the following (Borreani, 2025b).

5.3.1 Concept 1 – Evaluation

Concept 1 was taken as the reference in the Pugh Matrix. The concept is beneficial for the available payload, due to the extra frame being always installed when bodywork is required. The suspension of Concept 1, compared to most of the other concepts, requires fewer parts, which is why the total weight of the battery suspension would be lower. As a result, all the other concepts, besides Concept 7, perform worse in the evaluation compared to Concept 1. Regarding criteria 2-6, 8 and 9, the concept seems sufficient. When taking criteria 7 into consideration, problems may occur when trying to install bodywork. Do to the fact that the battery is on the same level as the bodybuilder frame, collisions may occur, which is a big weak point of the concept, due to the high customer importance ranking in the QFD. Normally, the installation of Scania components doesn't exceed the truck frame, so that the bodybuilder frame is free for bodywork installation.

5.3.2 Concept 2 – Evaluation

Concept 2 is less beneficial regarding criteria 1, due to the mentioned reasons in the Section 5.3.1. The addition of the double frame would increase the weight of the installation, compared to Concept 1. Criteria 2-5 and 8-9 are fulfilled in the same way as in Concept 1. Criteria 6 is lower, due to the extra costs which occur due to the need for a double frame. Since the battery suspension installation finishes with the height of the truck frame, criteria 7 is fulfilled better, because there is no potential for collisions with bodywork, such as in Concept 1. Therefore, Concept 2 and all following concepts fulfil criteria 7 better than Concept 1.

5.3.3 Concept 3 – Evaluation

Concept 3 offers a lower maximum payload, due to the mentioned reasons above. Furthermore, due to the usage of a crossmember between the LHS and RHS of the frame, the potential for collisions with other parts on the frame exists. Therefore, problems can occur when installing the battery suspension at different positions on the frame, which is why criteria 2 is less fulfilled. Nevertheless, the crossmember results in a more robust design, because it gives the frame more strength, which is why criteria 4 is fulfilled better in Concept 3.

5.3.4 Concept 4 – Evaluation

Concept 4 has the same evaluation as Concept 3, because the general design of both concepts is similar. The used crossmember in Concept 4 has a U-shape, instead of going at the top between the LHS and RHS of the frame, such as in Concept 2. Nevertheless, the resulting attributes of Concept 4 are the same as Concept 3 when compared to Concept 1.

5.3.5 Concept 5 – Evaluation

Concept 5 performs like the previous two concepts in the evaluation. Instead of the crossmember going from the LHS to the RHS of the frame, Concept 5 has the beam going between the sides. Therefore, the beam has the potential to collide with other components, which explains the evaluation of criteria 2. The reasoning for the other criteria are the same as given for the two concepts above.

5.3.6 Concept 6 – Evaluation

Concept 6 performs better in criteria 2 because it doesn't exceed the height of the truck frame, and at the same time, has a connection between the LHS and RHS of the frame, which goes underneath the frame, where the issue of potential collisions with other parts is lower. Furthermore, this connection results in more robustness of the concept, which is why criteria 4 is fulfilled better, compared to Concept 1.

5.3.7 Concept 7 – Evaluation

Concept 7 is the only concept which is evaluated the same regarding criteria 1. This is achieved because the additional installed parts for the battery suspension of Concept 7 is similar to Concept 1. Therefore, the difference between the weight of the installation of the battery suspensions of both concepts is similar. As a result, the maximum payload will be similar as well. The evaluation of Concept 7 regarding criteria 2 is better, because the battery suspension doesn't exceed the height of the truck frame, why a collision with bodywork is less likely than with Concept 1. Criteria 4 is evaluated better as well, because the design of the bracket, which connects from the top of the frame to the bottom of the bulkhead, results in a robust design and prevents the twisting of the battery below the frame. Furthermore, the concept is only suspended to one side of the frame, which is beneficial to achieve flexibility in positioning the battery suspension, and at the same time enables an unsymmetrical design, which were defined as the priorities in the Section 4.2.

5.3.8 Concept 8 – Evaluation

Concept 8 was evaluated better regarding criteria 4 because of the additional connection between the left and right sides of the battery suspension. Furthermore, as in the previous concepts, criteria 7 is evaluated better, because the suspension doesn't exceed the height of the truck frame, which is why collision with bodywork is less likely.

5.3.9 Result of the Evaluation

The result of the Pugh Matrix shows that the defined criteria, which are based on the QFD tool, are fulfilled the best with Concept 7. The concept performs strongly in the criteria regarding enabling different positions and bodywork, which are crucial for the design of the new battery suspension. Furthermore, due to the low amount of needed parts, it will be affordable and at the same time robust. Moreover, due to the installation only considering attachments on one side of the frame, where the battery will be mounted, the risk of possible collisions is low, and it allows the unsymmetrical positioning of the batteries to the frame. Additionally, this design allows the attachments at several positions on both sides of the frame, due to the stated argument before.

These mentioned aspects were evaluated as the most important functional requirements in the QFD, which is why Concept 7 will be further developed in the following. Therefore, the project team will optimise the concept, which aims to result in a more sufficient design of the battery suspension.

5.4 Concept Development

After the concept was selected, the available space was analysed in more detail using Catia V5. It became clear that the area inside the frame was more limited than initially expected, with several collision points that needed to be taken into consideration. The powertrain, along with cables, hoses and the air tanks, occupies a significant amount of space between the frame, see Figure 42. Because of this, the concept needed to be compact and avoid extending too far into the inside of the frame, which meant the design had to include a structure that angled downward relatively quickly. It also needed to avoid large components that might interfere with existing systems.

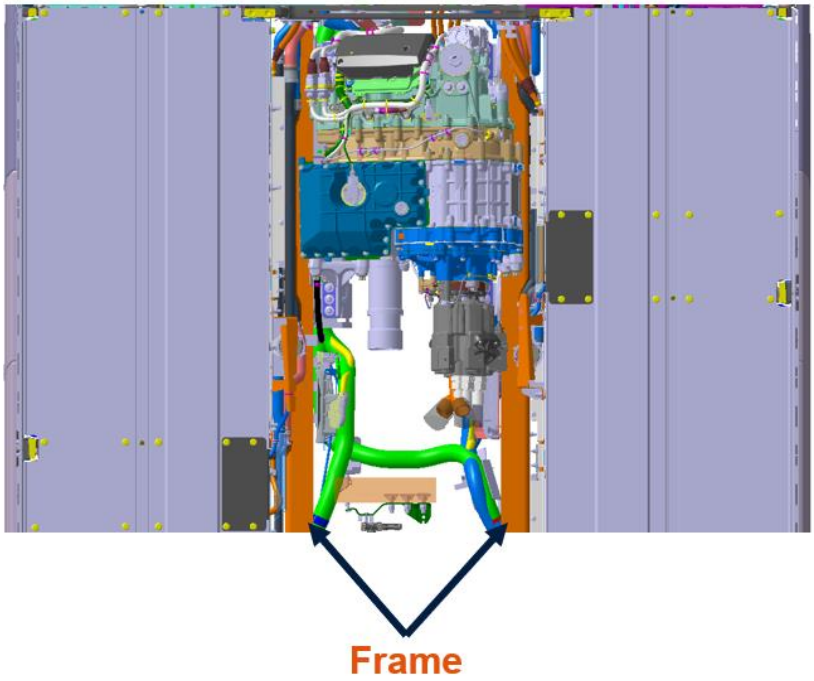


Figure 42: Potential Collision Points between the Frame

To address these challenges, the design was further developed to better fit the real conditions and ensure it could be integrated smoothly. During this stage, inspiration was drawn from the powertrain suspension, which had a structure well suited to the design goals. This became the starting point for refining the concept.

Figure 43 illustrates how the powertrain suspension is designed. The suspension starts in the inside of the frame and uses a bracket that goes down below the frame. This design remembers of the bracket design of Concept 7, see Section 5.2.7. Therefore, in consideration of the high potential of collisions inside the frame, the project team took the inspiration from the powertrain suspension.

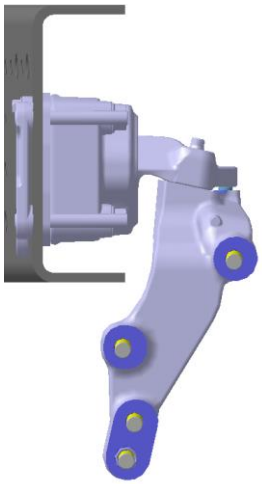


Figure 43: Powertrain Suspension

This inspiration guided the continued development of the concept, which can be seen from the same perspective in Figure 44.

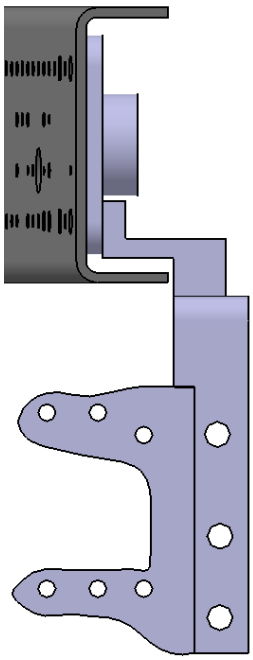


Figure 44: Bracket Design after the Inspiration

The completely developed concept consists of four main parts, as shown in Figure 45, which visualises the same part from two different perspectives.

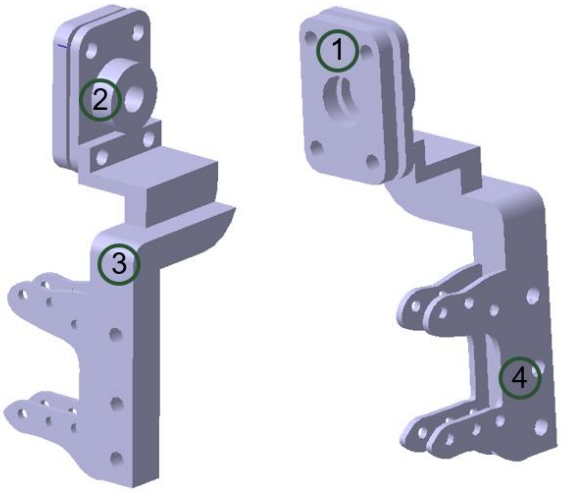


Figure 45: Bracket Design Concept 7

Part 1 and 2 are separated by the frame, which runs through both parts. The frame is hidden in Figure 45 to be able to see the newly designed parts. Part (1), see Figure 46, sits at the outside of the frame, where the clamping bracket and bulkhead are located. The spacer is required for the mounting of the clamping bracket. A similar part is used in the MS, which is why this design was taken over into the design of the new battery suspension. The two holes at the top and bottom of the part are aligned with the holes of the attachments of the clamping bracket. The central hole is designed to allow the beam to pass through it, which is why the part has the prepared interfaces for the connection to the clamping bracket and the beam.

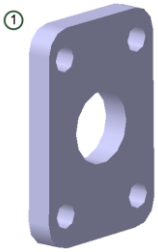


Figure 46: Spacer

Part (2) has a very similar design and function to part (1). The main difference lies in how the beam is secured as it passes through the central hole. In part (2), this hole is extruded, which helps to lock the beam firmly in place. The reason the suspension setup differs between the inside and outside of the frame is the difference in available space. There is more room on the outside, which allows for mounting the clamping bracket. On the inside, however, space is more limited. To address this, a new suspension system was

developed, one that requires less space but still keeps the beam stable and securely in place, just as it should. This part is illustrated in Figure 47.

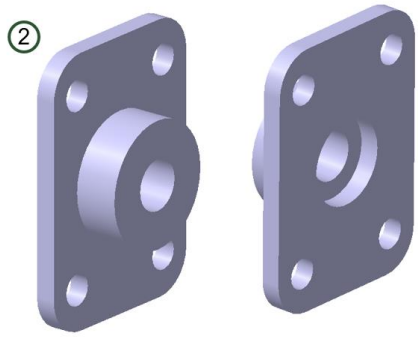


Figure 47: Bracket inside the Frame

Part (3) is an arm that connects to part (2) at the top and extends down to the bulkhead, where it is also attached. At the top of part (3), there are two holes that line up with the lower two holes of part (2). Screws are used here to firmly connect the two parts. From there, the part tries to go below the frame as fast as possible, to not collide with components in between the frame. Furthermore, the attachment points at the top don’t align with the bottom of the bulkhead, which is why the part has an L-shape (see Figure 48).

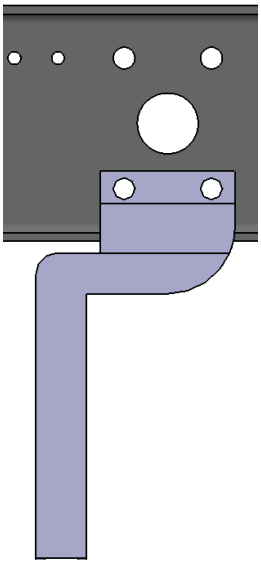


Figure 48: Bracket Arm L-Shape

The bottom of part (3) has a U-shaped design with six holes in total, three on each side. This shape allows it to slide into part (4), where it is then secured with screws. The part is shown in Figure 49.

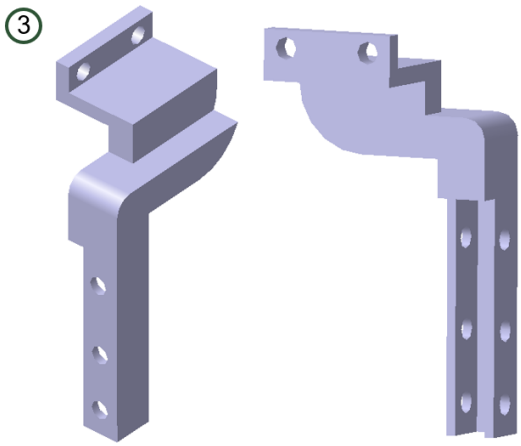


Figure 49: Bracket Arm

Part (4) connects part (3) to the lower Section of the bulkhead. It has a U-shaped design to ensure a proper and secure fit against the lower bulkhead connection. The component features a total of 15 holes distributed across its surfaces, three on each of the two vertical side plates, and additional holes on the slimmer, horizontal Section to allow attachment to part (3), see Figure 50.

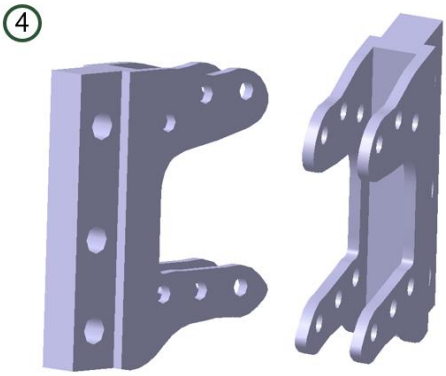


Figure 50: U-shape connection to the bottom of the Bulkhead

Following the concept of the battery suspension will be presented. The figures will show the left side of the battery suspension. Figure 51 shows the battery suspension when looking at it from the outside of the frame. The bulkhead, with the connection to the clamping bracket and beam, can be seen. Furthermore, the developed spacer between the frame and the clamping bracket is shown.

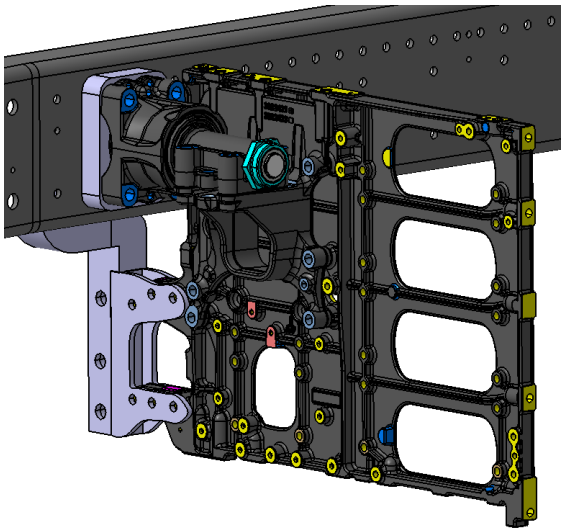


Figure 51: Battery Suspension Assembly outside the Frame

When looking from the inside of the frame, parts 2-4 can be seen. Figure 52 the connection between the developed parts, the frame and the bottom of the bulkhead is visible.

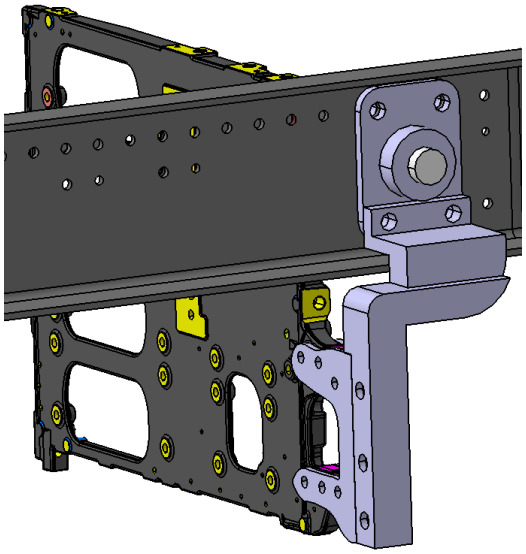


Figure 52: Battery Suspension Assembly inside the Frame

The feasibility of the design of the battery suspension will be analysed in the following Section. Therefore, the project team will conduct Generative Assembly Structural Analysis in Catia V5, to identify if the frame can handle the resulting stress.

Due to the learnings from the Section 5.1.4 showed that the resulting stress in the frame might be the limitation to suspend a battery suspension to one side of the frame, the following analyses will evaluate the resulting stress in the frame. The designed parts will not be included in the analysis because these parts are responsible for preventing the bending of the suspension underneath the frame. To identify if the frame can handle the stresses, a Generative Assembly Structural Analysis in Catia V5 will be conducted.

6.1 1st Catia Generative Assembly Structural Analysis

The design of Concept 7 consisted of using the bulkhead, clamping bracket, beam and the bracket arm, which is connected from the top of the frame to the bottom of the bulkhead. The concept didn't consider any reinforcement of the frame, such as crossmembers. Therefore, to first analyse if the frame can handle the stress, when installing an MP10 with two suspension points, an analysis in Catia V5 will be conducted. One frame member will be used and clamped at the front and back, where the axles of the truck are suspended. Furthermore, four virtual rigid parts will be created. Two virtual rigid parts simulate the clamping bracket and beam on the sides of the bulkheads. To these, the bulkheads will be connected. Moreover, to simulate the MP10, two virtual parts will be attached to the inside of the bulkheads and to the centre of gravity point of the MP10. On both virtual rigid parts, a load of 3500N will be applied in in Z-direction. This correlates to the total weight of the battery and suspension (700kg) and is multiplied by 10 times G. This is simulating the maximum loads, which occur during an urban application with good road conditions.

The results of the simulation can be seen in Figure 53. The analysis shows that in the top left hole in the frame, where the clamping bracket is mounted, a stress of over 666 MPa results. This value exceeds the maximum allowed stress Scania has defined internally. Due to the fact that the resulting stress exceeds the maximum value, the project team decided to conduct a second analysis, which includes changes in the concept.

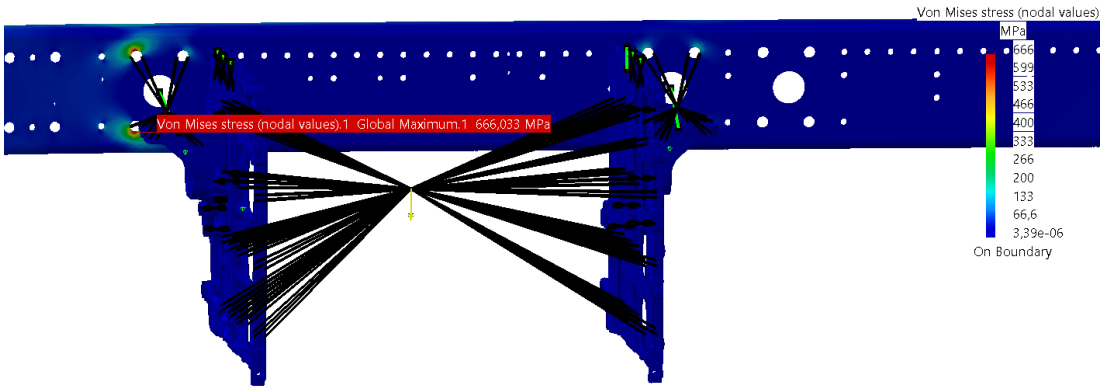


Figure 53: 1st Frame Analysis

6.2 2nd Catia Generative Assembly Structural Analysis

Following the results of the first analysis and inspired by the crossmember design of the MS, the project team decided to include three crossmembers at the same position, as they are used in the MS. As visible in Figure 54, one crossmember is positioned at the front, next to the first connection point of the battery suspension. The second crossmember is attached close behind the end of the battery suspension. The third crossmember is attached further to the back, behind the second crossmember. With the implementation of the three crossmembers, the project team wants to lower the stress in the frame holes. By implementing the crossmembers, the project team assumes that the stress will go through the crossmember to the other side member of the frame. As a result, the stress should be lower in the frame holes.

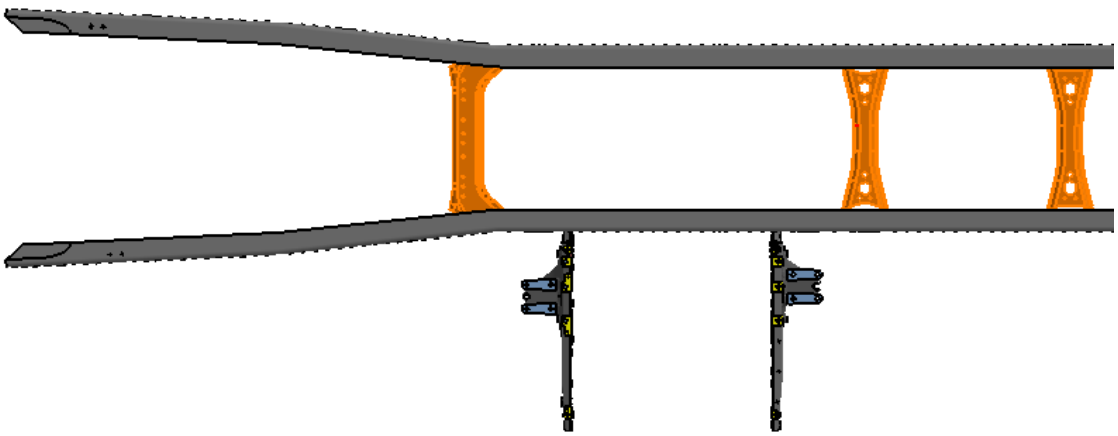


Figure 54: Three crossmembers implemented in the concept

The analysis setup is the same as before, with the only difference that the other side member of the frame and the three crossmembers are added. Between the frame and the crossmembers, rigid connection properties were defined, which simulate the fixation with bolts. The same loads are applied to the virtual rigid parts as in the first simulation.

As visible in Figure 55, the highest stress is 216 MPa, which doesn't exceed the maximum allowed value. The stress is located in one of the connection holes of the second crossmember. Besides that stress, the other holes of the crossmembers and holes where the battery suspension is suspended, have the highest stresses.

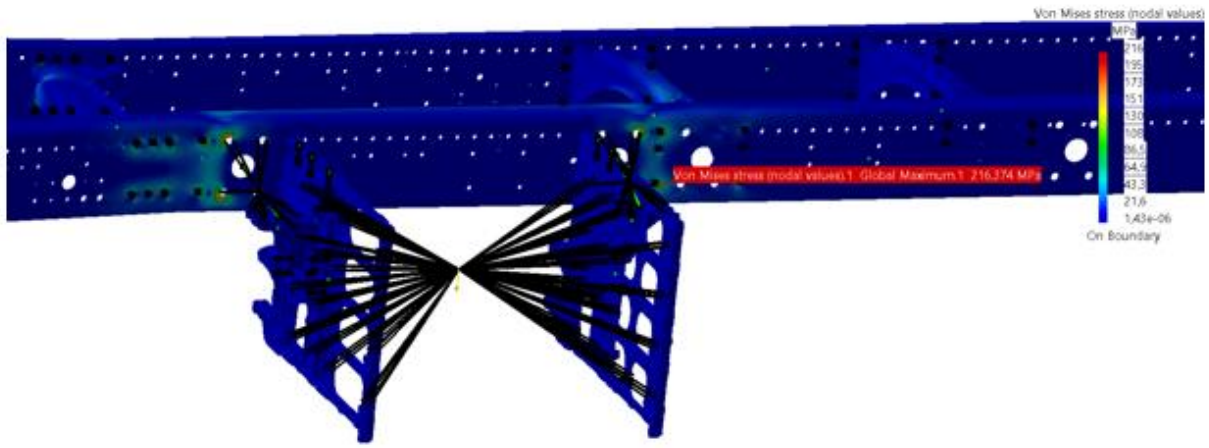


Figure 55: 2nd Frame Analysis

6.3 3rd Catia Generative Assembly Structural Analysis

A 3rd Generative Assembly Structural Analysis in Catia will be conducted, this time with two battery suspensions on one side of the frame. This analysis will be conducted to analyse how high the resulting stress in the frame is when suspending two battery packs on one side of the frame. This is of interest because it would increase the available battery capacity and increase the flexibility in positioning propulsion batteries to the frame.

The analysis setup is the same as before, with the only difference that a second battery suspension is installed next to the first one. The connection points and loads on the battery suspension are the same as described before.

The results of the analysis can be seen in Figure 56. The highest stress with 276 MPa is located in the same holes in the frame as in the previous simulation. Besides that, the highest stresses are again located in all other holes in the frame, where the battery suspension or crossmembers are located.

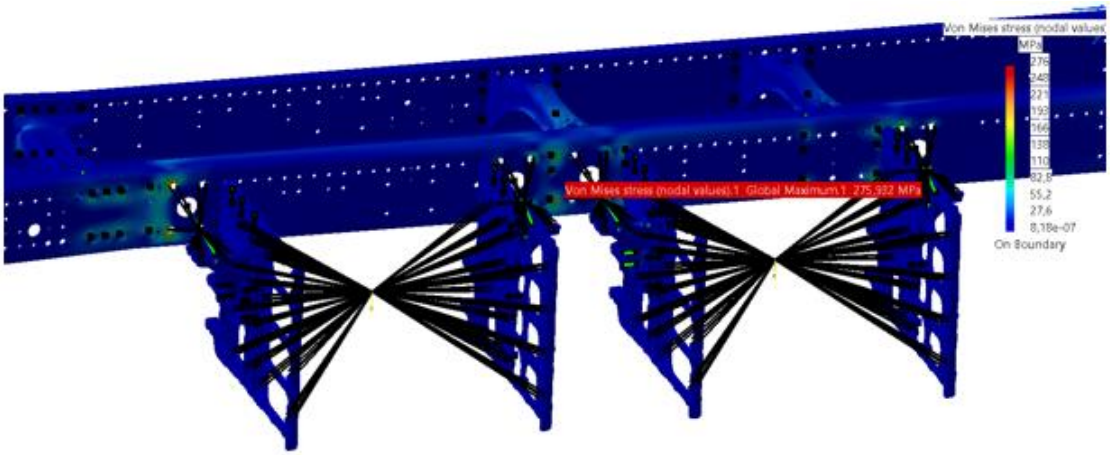


Figure 56: 3rd Frame Analysis

6.4 Conclusion of the Analyses

The conducted analyses brought the insides that a single suspended MP10 on one side of the frame is feasible. When using three crossmembers, in the original position of the MS, in combination with the bulkheads, the resulting stresses in the frame don’t exceed the maximum allowed value. Based on this analysis, suspending another MP10, at the same time, on the other side of the frame seems feasible as well. By that, the two MP10 batteries can be suspended symmetrical or unsymmetrical, which is visualised in Figure 57. Besides the two MP10 batteries, the future MP12, which is installed below the driver’s cab, is visualised as well in grey. In combination, a total battery capacity of 300 kWh is able to be achieved, which correlates with the target defined the QFD tool.

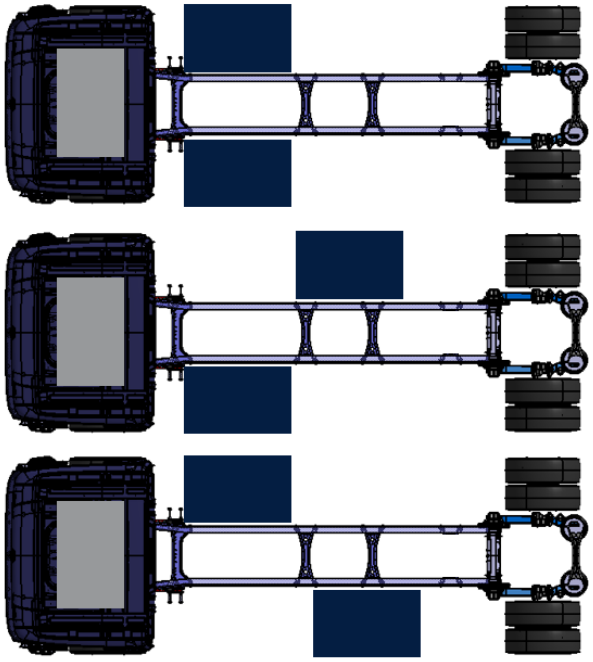


Figure 57: Possible Position of Propulsion Batteries

When suspending two MP10 on one side of the frame, the resulting stress also doesn’t exceed the maximum values, but due to the fact that the stress is almost exceeding the maximum value, further analyses are required to confirm the installation. The project team's concern is that further analyses, which consider the resulting stresses in the frame when the two suspended battery packs move in opposite directions, might be higher than the allowed maximum values. Nevertheless, this analysis shows that suspending an MP20 on one side of the frame, which has the same weight as suspending two MP10s, see the Section 2.1.2, seems promising. When suspending one MP20, the described problem of the opposite movement of the battery packs is not present. Therefore, further analyses regarding suspending an MP20 should be conducted.

7. Discussion

In the following Section, the project team will reflect on the outcome of the project. Furthermore, recommendations will be given, which future work Scania should apply to further develop the concept.

7.1 Reflection of the Project Outcome

The reflection of the project outcome will be conducted based on the deliverables, which were defined in Section 1.2.3:

Analysing the current design of the MS and understanding the reasons behind the design:

The analysis of the MS has shown that the focus during the development was an articulated truck (A4x2). Thereby, the installed battery capacity is crucial because it defines the range of the BEV truck. Since A-Trucks are mainly used for long-haulage transportations, the battery capacity aims to at least cover an operation of 4,5 hours, due to the legally defined break times for truck operators. Besides the battery capacity, the goal was to achieve a short axle distance, because a shorter truck is beneficial for the operation.

The design of the MS is based on the connection between the LHS and RHS of the chassis, with two beams running at the front and the back. The beams handle most of the load within the battery suspension. Due to the chosen design, the MS carries itself, which is why no additional reinforcement in the frame is required. This results in a lower truck weight, which is beneficial for the operation. Furthermore, suspending the MS on both sides of the frame ensures that the centre of gravity remains centrally aligned in the frame, which keeps the chassis horizontal and prevents it from twisting to one side.

A market analysis of how competitors design their battery suspensions and how they differ towards Scania's:

The conducted market analysis has shown that the design of the MS is unique on the market. All other competitors have single suspended propulsion batteries, which enables flexibility in changing the number and position of the batteries. Furthermore, competitors such as Mercedes-Benz and IVECO have an electric axle in the rear, which results in more free space for the installation of propulsion batteries. Regarding installed battery capacity, Scania is competitive compared to the rest of the market and, in most cases, ahead of its competitors.

Identifying space for the installation of propulsion batteries:

While the investigation it became clear that the current attachment of the MS, at the LHS and RHS of the frame, between the front and rear axle is beneficial for the installation of propulsion batteries. Therefore, the concept of the new battery suspension was developed specifically for this position.

The space between the frame is for Scania, not available, due to the positioning of the powertrain and propeller shaft. Moreover, below the driver's cab, Scania is already working on a solution, which is why this was not considered in the project. Lastly, due to the bodywork which is installed above the frame and because the truck dimensions are important, the propulsion batteries cannot be installed above the frame, or at the front or back of the chassis.

Concept generation for a new battery suspension:

With the support of the Pre-Development Work and the Concept Generation, a total of 8 different concepts for a new battery suspension were generated. With the support of the FMT method, the project team identified various solutions to address the challenge of reinforcing the frame. Furthermore, the developed concepts tried to fulfil the defined requirements of the QFD. In combination with the FMT, different ideas were generated, which had similarities but at the same time differed.

Concept evaluation and further developing the chosen concept:

Based on the Pugh Matrix, Concept 7 was evaluated as the chosen concept to be further developed. All concepts were compared with each other in how well they fulfil the defined customer requirements of the QFD. The chosen concept had the most points in the evaluation. Even though the result of the evaluation was close, the design of the Concept 7 enables the positioning at several positions on the frame and an unsymmetrical design. These aspects were defined as most crucial in the project targets, which is why the close advantage of the evaluation was sufficient to continue developing the concept further.

The further development of the chosen concept was inspired by the design of the powertrain suspension. As a result, the design of the concept changed, with the aim to go as fast down and below the frame member, to not collide with other components. The final design achieves this, but the design could be optimised so that it is more underneath the frame member

Analyses which consider the stress in the frame of the truck:

In total, three stress analyses of the frame were conducted with the help of the Generative Assembly Structural Analysis in Catia V5. The first analysis has shown that a single frame member can not handle the resulting stress by suspending one MP10 to it. By adding the second frame member and three crossmembers, which are in the original position of the MS, the resulting stress in the second analysis is below the maximum allowed value. Lastly, the third analysis simulated the stress in the frame when suspending two MP10 individually to the same side of the frame. The detail of the simulation was too simplified, which is why major conclusions couldn’t be made. Moreover, an analysis of suspending one MP20 to the side of the frame should have been conducted as well. This analysis would be of interest because it would increase the installed battery capacity on the truck.

Presenting a final concept which includes a new battery suspension, which enables more special customer applications:

The project team was able to present a final concept of a battery suspension. Due to being suspended on one side of the frame, without the need for a connection to the other side of the frame, the concept provides an opportunity to allow the realisation of more special customer applications. The defined bodyworks, which were identified during the project, are capable of being realised with the concept. Nevertheless, it needs to be analysed if the available battery capacities are sufficient for the individual customer's needs. Furthermore, the developed parts need to be analysed to determine if the parts fulfil the intended purpose, to prevent the twisting of the propulsion batteries below the frame. In comparison with the MS, the developed concept enables more special customer applications, due to achieving more flexibility in the positioning of the battery suspension.

The final concept is developed for the special order department within Scania, which seeks a concept which can be implemented quickly and cost-effectively:

All the generated concepts consider using three parts, which are used in the MS. By that, the concepts include developed and validated components, which increases the probability of an implementation of the concept in the future. By that, the required time and costs for the development of the concept are reduced. When considering the final concept, the mentioned aspects were achieved even better. The usage of the crossmembers in the original position of the MS further increases the similarities between the MS and the concept, which is beneficial for costs and time. The costs for the self-developed parts were not analysed, and on top of that, the parts were not analysed, which is why costs and time-effort can not be defined based on the conducted work.

7.2 Future Work

The conducted study of the project team clearly has shown that introducing a battery suspension, which is only suspended to one side of the frame, is feasible when considering one MP10 battery. Based on these insights, the project team is recommending the following future steps.

Firstly, the design of self-developed parts needs to be further developed. Therefore, analyses should be applied to see if the parts can handle the stress and fulfil the purpose. The goal of the design is that the parts prevent the twisting of the batteries below the frame. Based on these results, the dimensions and design of the self-developed parts will change.

Furthermore, an investigation should be started, which concentrates on finding possible collision points between the newly designed battery suspension and components, which are mounted inside the frame. On top of that, the described components in Section 2.1.3, which are connected to the MS, might require a new position or components such as hoses and cables need a new route along the frame to the battery suspension. Based on these insights, the design of the self-developed parts might further change. In combination with further detailed analysis, which considers additional dynamic scenarios, the suspension of one MP10 to the side of the frame can be further approved.

Additionally, further tests with two MP10 batteries on one frame side should be conducted to prove if the installation is feasible. The resulting dynamics, if the two suspended battery packs move in opposite directions, require further detailed analyses. Moreover, the same tests should be applied with one MP20 battery to see if that installation is feasible. Based on these results, further reinforcement might be required to handle the stresses in the frame.

Derived from these insights, further analyses can be conducted, where one MP10, two MP10 or one MP20 battery pack are suspended at different positions to the frame. Therefore, a variety of different battery layouts, which means different amounts of battery packs and positions to the frame, should be tested to evaluate which configurations are possible.

The mentioned future steps will help Scania to develop a solution for their customers, which enables the different customer applications with electric B-Trucks. The conducted work is a first step in fulfilling the customer needs, and when applying the future work, Scania will be able to fulfil the needs of their customers better in the future.

8. Conclusion

When evaluating the existing battery suspension system, it became clear that the Mega structure’s symmetrical design resulted in limitations on adaptability and modification. The system relies on both sides working together to balance and support the batteries, creating an interdependent and complex setup that lacks flexibility. This makes modifying the structure to an asymmetrical layout challenging. The Pre-Study has shown that the customers require flexibility in the position of the battery suspension, to enable different customer applications. To overcome the limitations of the MS, the development of a completely new battery suspension system was conducted.

The new battery suspension concept operates independently on each side of the frame, providing flexibility in layout and configuration. On both sides of the frame, the battery suspension can be moved forward and backwards along the straight running frame. As a result, customers have the option to choose different positions of the battery suspension, which can be used to maximise payload. Moreover, the gained free space enables the installation of bodywork. Due to the usage of similar parts between the MS and the newly developed battery suspension, the new designs becomes more affordable. On top of that, the usage of the developed and tested components ensures that the battery suspension is robust and safe at the same time. When suspending one MP10 each on the LHS and RHS of the frame, a total battery capacity of 200 kWh can be achieved. Combined with a battery pack below the driver's cab, which is in development at Scania, a total capacity of 300 kWh will be available. Lastly, due to considering the maximum truck dimensions and taking inspiration from the MS, the newly designed battery suspension is legal to operate. Therefore, the developed battery suspension fulfils the customer requirements, which were defined in the QFD tool.

This new concept of the battery suspension was designed for electric B-Trucks for Scania, and the project has shown that there are opportunities to optimise the fulfilment of the customer needs. In combination with the mentioned future steps, Scania can be able to develop a solution for the future, which fulfils the requirements regarding flexibility better.

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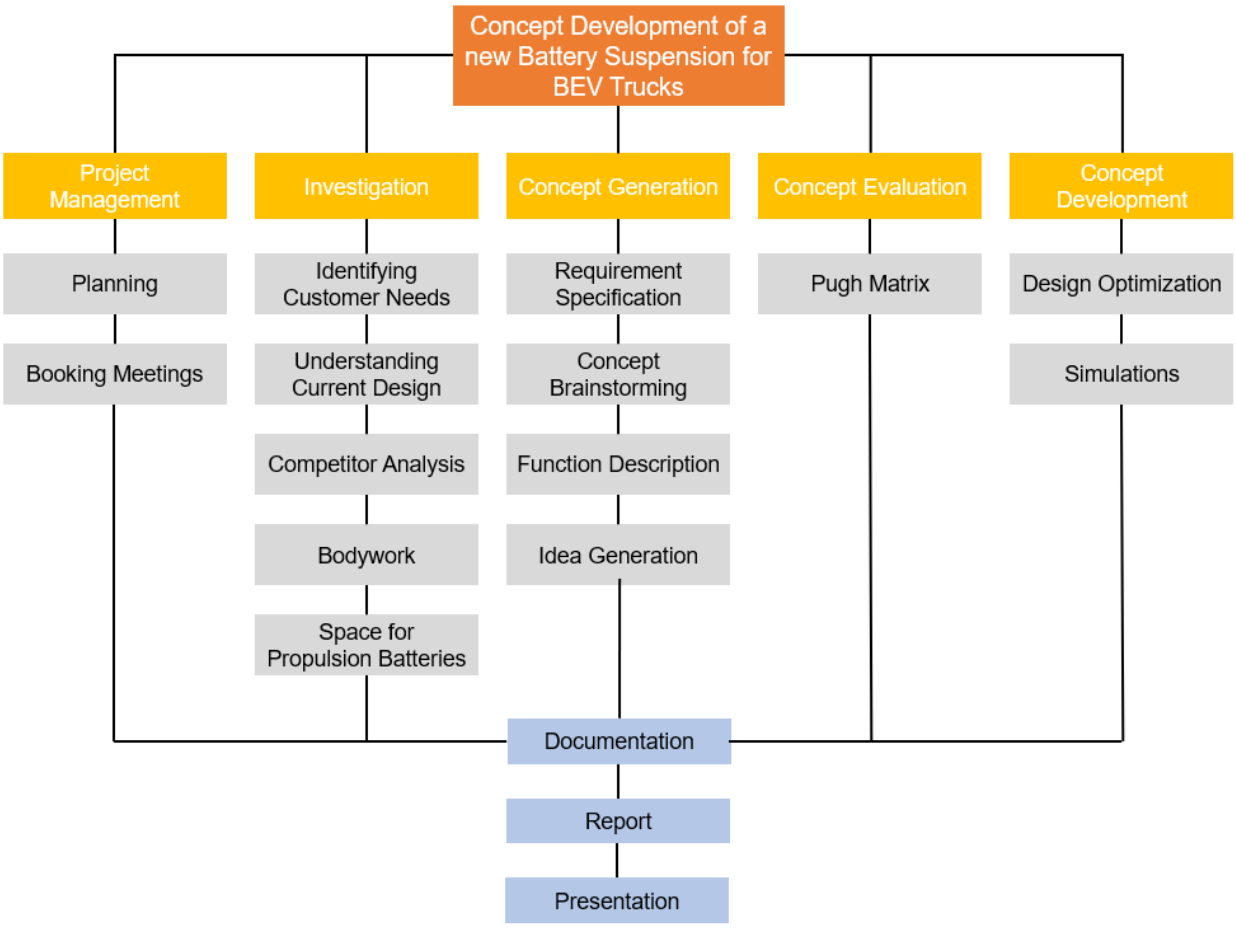
Appendix

Appendix A: WBS XX

Appendix B: Quality Function Deployment XXI

Appendix C: Function Means Tree XXII

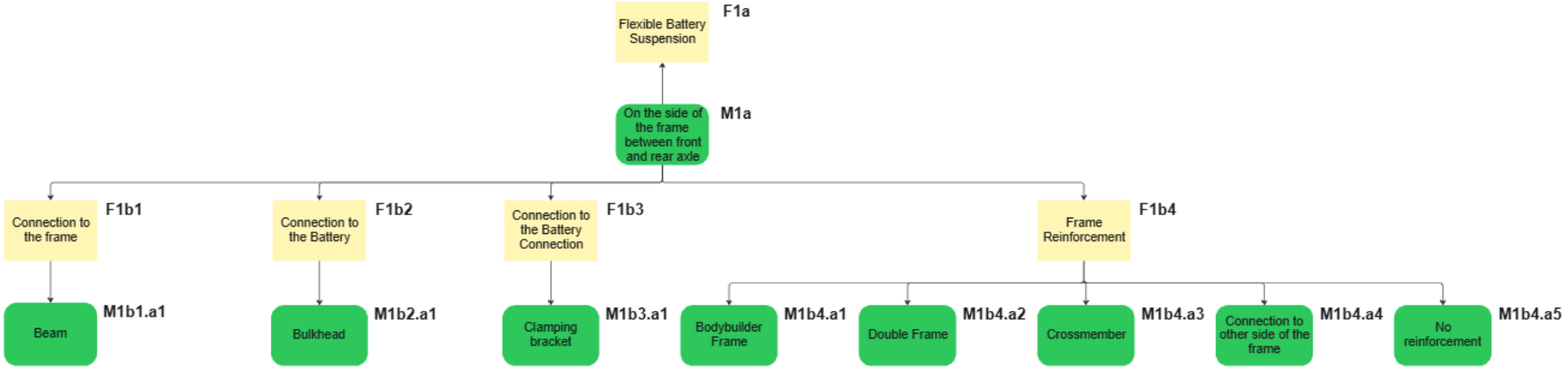
Appendix A: WBS



Appendix B: Quality Function Deployment

1: low, 5: high	Functional Requirements (How's) →	Minimum required reinforcement	Lowering the stress on the frame	Flexible attachment points on the frame	Up to 300 kWh battery capacity	Unsymmetrical battery design	Compact battery supension design	Safe storage of the batteries on the truck	Chassis width max. 2550 mm	Several attachment points for the batteries	Gross vehicle weight (GVW)
Customer importance rating	Customer Requirements - (What's) ↓										
4	Maximising payload	9	3	9	9	3	3		1	9	9
5	Different positions of the Battery Suspension	3	3	9	3	9	3	3		9	3
3	Different ranges (km)		3	3	9	9	3	3	3	3	3
3	Robust	9	9	3	3	9	1	9	3	3	1
4	Safe	9	9	1	1	3	1	9		3	9
3	Affordable	9	9	3	9	3	1	3		3	3
5	Enabling Bodybuilder adaptions	1	1	9	3	9	3	1		9	1
5	Free space on the frame			3	3	9	3	1	1	9	
5	Legal to operate	3	3					3	9		9
	Technical importance score	161	146	172	148	222	76	121	72	210	158
	Importance %	11%	10%	12%	10%	15%	5%	8%	5%	14%	11%
	Priorities rank	**4**	**7**	**3**	**6**	**1**	**9**	**8**	**10**	**2**	**5**

Appendix C: Function Means Tree



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